

DEVOPS INTERVIEW QUESTION BANK

520 scenario-based multiple-choice questions

Linux | Git | CI/CD | Docker | Kubernetes | Cloud | Monitoring | Networking |
Security | Architecture

Questions first. Complete answer key and explanations at the end.

How to Use This Book

- Use the question bank as a timed interview simulation or as a topic-by-topic diagnostic.
- For each question, select one answer only. Do not consult the answer key until the session is complete.
- Suggested timing: 60-90 seconds for foundational and intermediate questions; 2-3 minutes for advanced and expert scenarios.
- Recommended scoring: 80%+ strong interview readiness, 65-79% developing, 50-64% needs focused revision, below 50% rebuild fundamentals.

Domain	Questions
Linux	52
Git	52
CI/CD	52
Docker	52
Kubernetes	52
Cloud	52
Monitoring and Observability	52
Networking	52
Security	52
Architecture and Incident Response	52
Total	520

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Candidate Answer Sheet

Write A, B, C, or D beside each question number. This section intentionally contains no answers.

Questions 1-130

001 ____	002 ____	003 ____	004 ____	005 ____	006 ____	007 ____	008 ____	009 ____	010 ____
011 ____	012 ____	013 ____	014 ____	015 ____	016 ____	017 ____	018 ____	019 ____	020 ____
021 ____	022 ____	023 ____	024 ____	025 ____	026 ____	027 ____	028 ____	029 ____	030 ____
031 ____	032 ____	033 ____	034 ____	035 ____	036 ____	037 ____	038 ____	039 ____	040 ____
041 ____	042 ____	043 ____	044 ____	045 ____	046 ____	047 ____	048 ____	049 ____	050 ____
051 ____	052 ____	053 ____	054 ____	055 ____	056 ____	057 ____	058 ____	059 ____	060 ____
061 ____	062 ____	063 ____	064 ____	065 ____	066 ____	067 ____	068 ____	069 ____	070 ____
071 ____	072 ____	073 ____	074 ____	075 ____	076 ____	077 ____	078 ____	079 ____	080 ____
081 ____	082 ____	083 ____	084 ____	085 ____	086 ____	087 ____	088 ____	089 ____	090 ____
091 ____	092 ____	093 ____	094 ____	095 ____	096 ____	097 ____	098 ____	099 ____	100 ____
101 ____	102 ____	103 ____	104 ____	105 ____	106 ____	107 ____	108 ____	109 ____	110 ____
111 ____	112 ____	113 ____	114 ____	115 ____	116 ____	117 ____	118 ____	119 ____	120 ____
121 ____	122 ____	123 ____	124 ____	125 ____	126 ____	127 ____	128 ____	129 ____	130 ____

Questions 131-260

131 ____	132 ____	133 ____	134 ____	135 ____	136 ____	137 ____	138 ____	139 ____	140 ____
141 ____	142 ____	143 ____	144 ____	145 ____	146 ____	147 ____	148 ____	149 ____	150 ____
151 ____	152 ____	153 ____	154 ____	155 ____	156 ____	157 ____	158 ____	159 ____	160 ____
161 ____	162 ____	163 ____	164 ____	165 ____	166 ____	167 ____	168 ____	169 ____	170 ____
171 ____	172 ____	173 ____	174 ____	175 ____	176 ____	177 ____	178 ____	179 ____	180 ____
181 ____	182 ____	183 ____	184 ____	185 ____	186 ____	187 ____	188 ____	189 ____	190 ____
191 ____	192 ____	193 ____	194 ____	195 ____	196 ____	197 ____	198 ____	199 ____	200 ____
201 ____	202 ____	203 ____	204 ____	205 ____	206 ____	207 ____	208 ____	209 ____	210 ____
211 ____	212 ____	213 ____	214 ____	215 ____	216 ____	217 ____	218 ____	219 ____	220 ____
221 ____	222 ____	223 ____	224 ____	225 ____	226 ____	227 ____	228 ____	229 ____	230 ____
231 ____	232 ____	233 ____	234 ____	235 ____	236 ____	237 ____	238 ____	239 ____	240 ____
241 ____	242 ____	243 ____	244 ____	245 ____	246 ____	247 ____	248 ____	249 ____	250 ____
251 ____	252 ____	253 ____	254 ____	255 ____	256 ____	257 ____	258 ____	259 ____	260 ____

Questions 261-390

261 ____	262 ____	263 ____	264 ____	265 ____	266 ____	267 ____	268 ____	269 ____	270 ____
271 ____	272 ____	273 ____	274 ____	275 ____	276 ____	277 ____	278 ____	279 ____	280 ____

281 ____	282 ____	283 ____	284 ____	285 ____	286 ____	287 ____	288 ____	289 ____	290 ____
291 ____	292 ____	293 ____	294 ____	295 ____	296 ____	297 ____	298 ____	299 ____	300 ____
301 ____	302 ____	303 ____	304 ____	305 ____	306 ____	307 ____	308 ____	309 ____	310 ____
311 ____	312 ____	313 ____	314 ____	315 ____	316 ____	317 ____	318 ____	319 ____	320 ____
321 ____	322 ____	323 ____	324 ____	325 ____	326 ____	327 ____	328 ____	329 ____	330 ____
331 ____	332 ____	333 ____	334 ____	335 ____	336 ____	337 ____	338 ____	339 ____	340 ____
341 ____	342 ____	343 ____	344 ____	345 ____	346 ____	347 ____	348 ____	349 ____	350 ____
351 ____	352 ____	353 ____	354 ____	355 ____	356 ____	357 ____	358 ____	359 ____	360 ____
361 ____	362 ____	363 ____	364 ____	365 ____	366 ____	367 ____	368 ____	369 ____	370 ____
371 ____	372 ____	373 ____	374 ____	375 ____	376 ____	377 ____	378 ____	379 ____	380 ____
381 ____	382 ____	383 ____	384 ____	385 ____	386 ____	387 ____	388 ____	389 ____	390 ____

Questions 391-520

391 ____	392 ____	393 ____	394 ____	395 ____	396 ____	397 ____	398 ____	399 ____	400 ____
401 ____	402 ____	403 ____	404 ____	405 ____	406 ____	407 ____	408 ____	409 ____	410 ____
411 ____	412 ____	413 ____	414 ____	415 ____	416 ____	417 ____	418 ____	419 ____	420 ____
421 ____	422 ____	423 ____	424 ____	425 ____	426 ____	427 ____	428 ____	429 ____	430 ____
431 ____	432 ____	433 ____	434 ____	435 ____	436 ____	437 ____	438 ____	439 ____	440 ____
441 ____	442 ____	443 ____	444 ____	445 ____	446 ____	447 ____	448 ____	449 ____	450 ____
451 ____	452 ____	453 ____	454 ____	455 ____	456 ____	457 ____	458 ____	459 ____	460 ____
461 ____	462 ____	463 ____	464 ____	465 ____	466 ____	467 ____	468 ____	469 ____	470 ____
471 ____	472 ____	473 ____	474 ____	475 ____	476 ____	477 ____	478 ____	479 ____	480 ____
481 ____	482 ____	483 ____	484 ____	485 ____	486 ____	487 ____	488 ____	489 ____	490 ____
491 ____	492 ____	493 ____	494 ____	495 ____	496 ____	497 ____	498 ____	499 ____	500 ____
501 ____	502 ____	503 ____	504 ____	505 ____	506 ____	507 ____	508 ____	509 ____	510 ____
511 ____	512 ____	513 ____	514 ____	515 ____	516 ____	517 ____	518 ____	519 ____	520 ____

Linux

LINUX

Questions 001-052 | 52 questions

Linux Questions

Q001**INTERMEDIATE**

A server shows load 24, CPU idle 55%, wa 35%, and many tasks in D state. What is the most likely cause?

- A. active swapping caused by memory pressure
- B. a DNS resolver failure
- C. storage I/O contention
- D. CPU saturation from too many runnable tasks

Q002**ADVANCED**

A server shows load 18, CPU idle 60%, vmstat shows b=16 and bi=80000. What is the most likely cause?

- A. CPU saturation from too many runnable tasks
- B. active swapping caused by memory pressure
- C. a DNS resolver failure
- D. blocked disk reads

Q003**ADVANCED**

A server shows load 30 on a 16-core host, CPUs mostly idle, NFS latency is 900 ms. What is the most likely cause?

- A. CPU saturation from too many runnable tasks
- B. a DNS resolver failure
- C. slow NFS causing uninterruptible waits
- D. active swapping caused by memory pressure

Q004**EXPERT**

A server shows load rises after a SAN path degrades while CPU use remains below 30%. What is the most likely cause?

- A. CPU saturation from too many runnable tasks
- B. a DNS resolver failure
- C. active swapping caused by memory pressure
- D. storage latency

Q005

INTERMEDIATE

Which interpretation is most accurate when MemFree is 150 MB, MemAvailable is 11 GB, Cached is 9 GB, and swap use is zero?

- A. The host is critically out of memory solely because MemFree is low
- B. Cached memory is unreclaimable until reboot
- C. The host still has substantial reclaimable memory
- D. The CPU scheduler is malfunctioning

Q006

INTERMEDIATE

Which interpretation is most accurate when free reports 95% used, but buff/cache is 22 GB on a 32 GB host?

- A. Cached memory is unreclaimable until reboot
- B. The host is critically out of memory solely because MemFree is low
- C. The CPU scheduler is malfunctioning
- D. Linux is using spare RAM for cache

Q007

ADVANCED

Which interpretation is most accurate when MemAvailable drops below 400 MB and major page faults climb sharply?

- A. Cached memory is unreclaimable until reboot
- B. The host is critically out of memory solely because MemFree is low
- C. The host is under real memory pressure
- D. The CPU scheduler is malfunctioning

Q008

EXPERT

Which interpretation is most accurate when swap-in and swap-out remain high for ten minutes while latency rises?

- A. The CPU scheduler is malfunctioning
- B. Cached memory is unreclaimable until reboot
- C. The host is critically out of memory solely because MemFree is low
- D. active swapping is degrading performance

Q009

INTERMEDIATE

In Linux, a process is marked Z in ps. What is the correct conclusion or action?

- A. The filesystem must be remounted
- B. Its parent has not collected its exit status
- C. The process is consuming CPU in the background
- D. The kernel thread must be killed with SIGKILL

Q010**ADVANCED**

In Linux, hundreds of defunct child processes accumulate under one PID. What is the correct conclusion or action?

- A. Fix or restart the parent process
- B. The filesystem must be remounted
- C. The kernel thread must be killed with SIGKILL
- D. The process is consuming CPU in the background

Q011**ADVANCED**

In Linux, kill -9 on a zombie has no effect. What is the correct conclusion or action?

- A. The process is consuming CPU in the background
- B. The process is already dead
- C. The filesystem must be remounted
- D. The kernel thread must be killed with SIGKILL

Q012**EXPERT**

In Linux, PID 1 adopts orphaned children after their parent exits. What is the correct conclusion or action?

- A. The kernel thread must be killed with SIGKILL
- B. The process is consuming CPU in the background
- C. The filesystem must be remounted
- D. PID 1 should reap them

Q013**INTERMEDIATE**

A filesystem issue is observed: df -h shows 45% used but new files cannot be created. What is the best response?

- A. Run fsck immediately on the mounted filesystem
- B. Restart the network service
- C. Check inode usage with df -i
- D. Increase the filesystem block size without cleanup

Q014**INTERMEDIATE**

A filesystem issue is observed: df -i shows IUse% at 100%. What is the best response?

- A. Restart the network service
- B. Increase the filesystem block size without cleanup
- C. Remove or consolidate many small files
- D. Run fsck immediately on the mounted filesystem

Q015

ADVANCED

A filesystem issue is observed: an application creates millions of tiny cache files. What is the best response?

- A. Increase the filesystem block size without cleanup
- B. Restart the network service
- C. inode exhaustion may occur before space exhaustion
- D. Run fsck immediately on the mounted filesystem

Q016

EXPERT

A filesystem issue is observed: /var is full by inode count due to session files. What is the best response?

- A. Run fsck immediately on the mounted filesystem
- B. Increase the filesystem block size without cleanup
- C. Restart the network service
- D. clean old session files and add retention controls

Q017

FOUNDATIONAL

Given that a script is mode 0644 and returns Permission denied when executed, what is the best solution or interpretation?

- A. Run the process as root permanently
- B. Disable mandatory access control globally
- C. Grant mode 0777 to the entire parent tree
- D. add the execute bit

Q018

INTERMEDIATE

Given that a directory is readable but lacks execute permission, what is the best solution or interpretation?

- A. Grant mode 0777 to the entire parent tree
- B. users cannot traverse it
- C. Run the process as root permanently
- D. Disable mandatory access control globally

Q019

ADVANCED

Given that a shared directory needs new files to inherit its group, what is the best solution or interpretation?

- A. Grant mode 0777 to the entire parent tree
- B. Run the process as root permanently
- C. Disable mandatory access control globally
- D. set the setgid bit on the directory

Q020**EXPERT**

Given that a process must bind to port 80 without running fully as root, what is the best solution or interpretation?

- A. Run the process as root permanently
- B. Grant mode 0777 to the entire parent tree
- C. Disable mandatory access control globally
- D. grant only CAP_NET_BIND_SERVICE

Q021**FOUNDATIONAL**

A systemd service has this issue: systemctl shows failed with exit-code. What should you do first?

- A. Delete and recreate the unit file without inspecting logs
- B. Add an arbitrary sleep before startup
- C. Reboot after every failed start
- D. inspect journalctl -u for the unit

Q022**INTERMEDIATE**

A systemd service has this issue: a service works manually but fails under systemd due to a missing environment variable. What should you do first?

- A. Delete and recreate the unit file without inspecting logs
- B. declare the variable in Environment or EnvironmentFile
- C. Reboot after every failed start
- D. Add an arbitrary sleep before startup

Q023**ADVANCED**

A systemd service has this issue: a unit starts before the network is actually usable. What should you do first?

- A. order it after network-online.target and enable the wait-online service where appropriate
- B. Reboot after every failed start
- C. Delete and recreate the unit file without inspecting logs
- D. Add an arbitrary sleep before startup

Q024

EXPERT

A systemd service has this issue: a daemon forks but the unit is configured Type=simple. What should you do first?

- A. Reboot after every failed start
- B. use the correct service Type or run the daemon in foreground
- C. Delete and recreate the unit file without inspecting logs
- D. Add an arbitrary sleep before startup

Q025

FOUNDATIONAL

What is the correct action when you want a process to shut down gracefully?

- A. Send SIGSTOP to request graceful cleanup
- B. Change the process nice value
- C. send SIGTERM
- D. Close the listening firewall port

Q026

INTERMEDIATE

What is the correct action when a process ignores SIGTERM and must be stopped?

- A. use SIGKILL as a last resort
- B. Send SIGSTOP to request graceful cleanup
- C. Change the process nice value
- D. Close the listening firewall port

Q027

ADVANCED

What is the correct action when a daemon supports configuration reload without restart?

- A. Close the listening firewall port
- B. Send SIGSTOP to request graceful cleanup
- C. Change the process nice value
- D. send SIGHUP if documented

Q028

EXPERT

What is the correct action when a container exits slowly because its shell wrapper does not forward signals?

- A. Close the listening firewall port
- B. Send SIGSTOP to request graceful cleanup
- C. use exec so the application becomes PID 1
- D. Change the process nice value

Q029

INTERMEDIATE

A file-descriptor problem occurs: an application reports Too many open files. What is the best diagnosis or action?

- A. check per-process and system file descriptor limits
- B. Clear the filesystem page cache
- C. Increase CPU frequency
- D. Disable TCP keepalive system-wide

Q030

ADVANCED

A file-descriptor problem occurs: ulimit -n is 1024 for a high-connection proxy. What is the best diagnosis or action?

- A. Clear the filesystem page cache
- B. raise the limit through the service manager and verify application settings
- C. Increase CPU frequency
- D. Disable TCP keepalive system-wide

Q031

ADVANCED

A file-descriptor problem occurs: lsof shows thousands of sockets in CLOSE_WAIT owned by one process. What is the best diagnosis or action?

- A. Clear the filesystem page cache
- B. Disable TCP keepalive system-wide
- C. the application is not closing sockets after peer shutdown
- D. Increase CPU frequency

Q032

EXPERT

A file-descriptor problem occurs: a leak steadily increases open descriptors until failure. What is the best diagnosis or action?

- A. Disable TCP keepalive system-wide
- B. Increase CPU frequency
- C. identify descriptor types with lsof and fix the application leak
- D. Clear the filesystem page cache

Q033

INTERMEDIATE

Disk usage behaves unexpectedly because df shows /var full but du totals far less. What is correct?

- A. du is always authoritative and df is incorrect
- B. Reformat the filesystem
- C. Increase the inode limit only
- D. look for deleted but open files with lsof +L1

Q034

INTERMEDIATE

Disk usage behaves unexpectedly because a huge log was deleted while the writer stayed running. What is correct?

- A. Reformat the filesystem
- B. du is always authoritative and df is incorrect
- C. restart or signal the writer to reopen the log
- D. Increase the inode limit only

Q035

ADVANCED

Disk usage behaves unexpectedly because logrotate uses copytruncate for a busy application. What is correct?

- A. Reformat the filesystem
- B. Increase the inode limit only
- C. du is always authoritative and df is incorrect
- D. it avoids reopen requirements but can lose a small amount of data

Q036

EXPERT

Disk usage behaves unexpectedly because a service supports USR1 to reopen logs. What is correct?

- A. Increase the inode limit only
- B. du is always authoritative and df is incorrect
- C. rotate the file and signal the service
- D. Reformat the filesystem

Q037

FOUNDATIONAL

A scheduled task has this requirement or failure: a script works interactively but fails from cron because a command is not found. What is the best answer?

- A. Use relative command paths because cron starts in the script directory
- B. use absolute paths or define PATH
- C. Disable time synchronization
- D. Cron always loads the user's interactive shell profile

Q038

INTERMEDIATE

A scheduled task has this requirement or failure: a cron job uses relative paths and writes files in the wrong place. What is the best answer?

- A. Disable time synchronization
- B. Cron always loads the user's interactive shell profile
- C. Use relative command paths because cron starts in the script directory
- D. set an explicit working directory

Q039

ADVANCED

A scheduled task has this requirement or failure: a cron job needs credentials from a shell profile. What is the best answer?

- A. Use relative command paths because cron starts in the script directory
- B. Disable time synchronization
- C. Cron always loads the user's interactive shell profile
- D. load them explicitly from a protected file or secret mechanism

Q040

ADVANCED

A scheduled task has this requirement or failure: a monthly job should run at 02:00 on the first day. What is the best answer?

- A. use 0 2 1 * *
- B. Disable time synchronization
- C. Use relative command paths because cron starts in the script directory
- D. Cron always loads the user's interactive shell profile

Q041

FOUNDATIONAL

During Linux network troubleshooting, you need to identify which process listens on TCP 8080. What is the correct conclusion or next step?

- A. Flush the Git credential cache
- B. Increase the process memory limit
- C. The service must be converted to UDP
- D. use ss -ltnp

Q042

INTERMEDIATE

During Linux network troubleshooting, a service listens only on 127.0.0.1. What is the correct conclusion or next step?

- A. remote clients cannot connect directly
- B. The service must be converted to UDP
- C. Flush the Git credential cache
- D. Increase the process memory limit

Q043

ADVANCED

During Linux network troubleshooting, SYN packets arrive but no SYN-ACK leaves. What is the correct conclusion or next step?

- A. Increase the process memory limit
- B. check local firewall and whether a process is listening
- C. The service must be converted to UDP
- D. Flush the Git credential cache

Q044

EXPERT

During Linux network troubleshooting, connections work locally but time out remotely. What is the correct conclusion or next step?

- A. Increase the process memory limit
- B. compare bind address, firewall, routing, and security-group rules
- C. Flush the Git credential cache
- D. The service must be converted to UDP

Q045

INTERMEDIATE

For LVM storage, a volume group has free extents and /data must grow. What is the correct action?

- A. extend the logical volume and then the filesystem
- B. Remove the physical volume first
- C. Format the logical volume after extending it
- D. Shrink the logical volume before the filesystem

Q046

INTERMEDIATE

For LVM storage, an ext4 logical volume was enlarged but df shows no change. What is the correct action?

- A. run resize2fs on the filesystem
- B. Shrink the logical volume before the filesystem
- C. Remove the physical volume first
- D. Format the logical volume after extending it

Q047

ADVANCED

For LVM storage, an XFS filesystem must be expanded. What is the correct action?

- A. Shrink the logical volume before the filesystem
- B. Remove the physical volume first
- C. Format the logical volume after extending it
- D. use xfs_growfs while mounted

Q048

EXPERT

For LVM storage, you plan to shrink a filesystem. What is the correct action?

- A. verify filesystem support and shrink the filesystem before the LV
- B. Remove the physical volume first
- C. Format the logical volume after extending it
- D. Shrink the logical volume before the filesystem

Q049

INTERMEDIATE

A virtual Linux host shows this behavior: top shows high st on a virtual machine. What does it indicate?

- A. The filesystem journal is full
- B. the hypervisor is not scheduling the VM enough CPU
- C. The guest kernel has an inode leak
- D. The DNS resolver is consuming CPU

Q050

ADVANCED

A virtual Linux host shows this behavior: application latency tracks CPU steal spikes. What does it indicate?

- A. The filesystem journal is full
- B. The DNS resolver is consuming CPU
- C. investigate host contention or move the VM
- D. The guest kernel has an inode leak

Q051

ADVANCED

A virtual Linux host shows this behavior: CPU user and system are low but run queue is high and steal is 40%. What does it indicate?

- A. The guest kernel has an inode leak
- B. the VM is CPU-starved by the host
- C. The DNS resolver is consuming CPU
- D. The filesystem journal is full

Q052**EXPERT**

A virtual Linux host shows this behavior: a noisy neighbor affects a shared cloud VM. What does it indicate?

- A. The DNS resolver is consuming CPU
- B. The filesystem journal is full
- C. choose a less contended host or dedicated capacity
- D. The guest kernel has an inode leak

Git

GIT

Questions 053-104 | 52 questions

Git Questions

Q053**INTERMEDIATE**

In Git, git reset --hard HEAD~3 removed three unpushed commits. What is the best answer?

- A. Use git revert on the current HEAD only
- B. Run git pull --rebase from the remote
- C. Delete the repository and clone again
- D. use git reflog and create a branch at the old HEAD

Q054**ADVANCED**

In Git, a branch pointer was moved backward accidentally. What is the best answer?

- A. Use git revert on the current HEAD only
- B. Delete the repository and clone again
- C. inspect git reflog
- D. Run git pull --rebase from the remote

Q055**ADVANCED**

In Git, a commit is unreachable but garbage collection has not run. What is the best answer?

- A. Use git revert on the current HEAD only
- B. recover it from reflog or fsck and attach a reference
- C. Delete the repository and clone again
- D. Run git pull --rebase from the remote

Q056**EXPERT**

In Git, a developer deleted a local branch containing unique commits. What is the best answer?

- A. Delete the repository and clone again
- B. Run git pull --rebase from the remote
- C. Use git revert on the current HEAD only
- D. find the old tip in reflog and recreate the branch

Q057**INTERMEDIATE**

In Git, a rebased private branch is rejected on push. What is the best answer?

- A. Use --force and ignore remote changes
- B. Merge the remote branch after the rebase without review
- C. Use a normal push without changing history
- D. git push --force-with-lease

Q058

ADVANCED

In Git, you must replace remote history but avoid clobbering a teammate's new commit. What is the best answer?

- A. Use --force and ignore remote changes
- B. use --force-with-lease after fetching
- C. Use a normal push without changing history
- D. Merge the remote branch after the rebase without review

Q059

ADVANCED

In Git, git push --force would work but is risky. What is the best answer?

- A. prefer --force-with-lease
- B. Use a normal push without changing history
- C. Merge the remote branch after the rebase without review
- D. Use --force and ignore remote changes

Q060

EXPERT

In Git, a shared branch was rebased locally. What is the best answer?

- A. Use --force and ignore remote changes
- B. Use a normal push without changing history
- C. Merge the remote branch after the rebase without review
- D. coordinate first and avoid rewriting shared history

Q061

INTERMEDIATE

In Git, you want a linear local feature history before integration. What is the best answer?

- A. rebase the feature branch onto the updated main branch
- B. Always reset main to the feature branch
- C. Delete merge commits from all shared branches
- D. Cherry-pick every file manually

Q062

ADVANCED

In Git, you need to preserve the exact topology of a collaborative branch. What is the best answer?

- A. Always reset main to the feature branch
- B. merge rather than rebase
- C. Delete merge commits from all shared branches
- D. Cherry-pick every file manually

Q063**ADVANCED**

In Git, a merge commit records when two lines of development joined. What is the best answer?

- A. this is useful for auditability
- B. Always reset main to the feature branch
- C. Cherry-pick every file manually
- D. Delete merge commits from all shared branches

Q064**EXPERT**

In Git, interactive rebase is used before publishing. What is the best answer?

- A. Cherry-pick every file manually
- B. Delete merge commits from all shared branches
- C. Always reset main to the feature branch
- D. it can squash, reorder, or edit private commits

Q065**INTERMEDIATE**

In Git, HEAD points directly to a commit instead of a branch. What is the best answer?

- A. Delete the detached commits before switching
- B. Push HEAD without creating a branch
- C. you are in detached HEAD state
- D. Run git gc to attach commits to a branch

Q066**ADVANCED**

In Git, you made useful commits while detached. What is the best answer?

- A. Push HEAD without creating a branch
- B. Run git gc to attach commits to a branch
- C. Delete the detached commits before switching
- D. create a branch before switching away

Q067**ADVANCED**

In Git, checking out a tag detaches HEAD. What is the best answer?

- A. Push HEAD without creating a branch
- B. this is normal for inspecting a fixed release
- C. Delete the detached commits before switching
- D. Run git gc to attach commits to a branch

Q068**EXPERT**

In Git, you want to experiment from an old commit. What is the best answer?

- A. create a new branch at that commit
- B. Run git gc to attach commits to a branch
- C. Push HEAD without creating a branch
- D. Delete the detached commits before switching

Q069**INTERMEDIATE**

In Git, a regression exists somewhere between a known good and bad commit. What is the best answer?

- A. Use git blame on every file
- B. use git bisect
- C. Inspect commits sequentially from oldest to newest
- D. Revert half the repository manually

Q070**ADVANCED**

In Git, tests can determine good or bad automatically. What is the best answer?

- A. Inspect commits sequentially from oldest to newest
- B. Use git blame on every file
- C. Revert half the repository manually
- D. run git bisect with an automated test script

Q071**ADVANCED**

In Git, the history contains 1024 candidate commits. What is the best answer?

- A. Inspect commits sequentially from oldest to newest
- B. Revert half the repository manually
- C. Use git blame on every file
- D. bisect needs roughly ten test steps

Q072**EXPERT**

In Git, a bisect session is complete. What is the best answer?

- A. Revert half the repository manually
- B. Use git blame on every file
- C. run git bisect reset
- D. Inspect commits sequentially from oldest to newest

Q073

INTERMEDIATE

In Git, a bad commit is already on a shared branch. What is the best answer?

- A. Delete the bad commit object
- B. use git revert
- C. Rewrite the public branch with reset --hard
- D. Create an unrelated empty commit

Q074

ADVANCED

In Git, you must revert a merge commit. What is the best answer?

- A. specify the mainline parent with -m
- B. Create an unrelated empty commit
- C. Rewrite the public branch with reset --hard
- D. Delete the bad commit object

Q075

ADVANCED

In Git, reset is proposed for a public production branch. What is the best answer?

- A. prefer revert to preserve published history
- B. Rewrite the public branch with reset --hard
- C. Delete the bad commit object
- D. Create an unrelated empty commit

Q076

EXPERT

In Git, a revert itself caused problems. What is the best answer?

- A. Create an unrelated empty commit
- B. Delete the bad commit object
- C. revert the revert
- D. Rewrite the public branch with reset --hard

Q077

INTERMEDIATE

In Git, you need to switch branches with uncommitted work. What is the best answer?

- A. Copy only the .git directory elsewhere
- B. Use git clean -fd to save the changes
- C. use git stash push
- D. Commit temporary work directly to main

Q078

ADVANCED

In Git, untracked files must also be stashed. What is the best answer?

- A. Copy only the .git directory elsewhere
- B. Commit temporary work directly to main
- C. use git stash push -u
- D. Use git clean -fd to save the changes

Q079

ADVANCED

In Git, you want to apply a stash but keep it for reuse. What is the best answer?

- A. Commit temporary work directly to main
- B. Use git clean -fd to save the changes
- C. Copy only the .git directory elsewhere
- D. use git stash apply

Q080

EXPERT

In Git, a stash conflicts when applied. What is the best answer?

- A. Commit temporary work directly to main
- B. Copy only the .git directory elsewhere
- C. resolve conflicts and stage the resolved files
- D. Use git clean -fd to save the changes

Q081

INTERMEDIATE

In Git, one specific commit from another branch is needed. What is the best answer?

- A. Reset the destination branch to the source
- B. Merge the entire source branch
- C. Copy the commit hash into a tag only
- D. use git cherry-pick

Q082

ADVANCED

In Git, a cherry-pick conflicts. What is the best answer?

- A. Reset the destination branch to the source
- B. resolve, stage, then run git cherry-pick --continue
- C. Merge the entire source branch
- D. Copy the commit hash into a tag only

Q083**ADVANCED****In Git, you picked the wrong commit and have not finished. What is the best answer?**

- A. Reset the destination branch to the source
- B. Merge the entire source branch
- C. run git cherry-pick --abort
- D. Copy the commit hash into a tag only

Q084**EXPERT****In Git, several adjacent commits are needed. What is the best answer?**

- A. Copy the commit hash into a tag only
- B. cherry-pick an explicit commit range in the required order
- C. Reset the destination branch to the source
- D. Merge the entire source branch

Q085**INTERMEDIATE****In Git, a merge conflict file contains ours and theirs markers. What is the best answer?**

- A. Delete both conflicted versions
- B. edit the file to the intended result and git add it
- C. Mark the file resolved without reviewing content
- D. Run git pull repeatedly until Git chooses

Q086**ADVANCED****In Git, git status says both modified. What is the best answer?**

- A. Delete both conflicted versions
- B. Run git pull repeatedly until Git chooses
- C. Mark the file resolved without reviewing content
- D. the same path changed incompatibly on both sides

Q087**ADVANCED****In Git, you want the version from the branch being merged in during a merge. What is the best answer?**

- A. Mark the file resolved without reviewing content
- B. Delete both conflicted versions
- C. choose theirs carefully and verify semantics
- D. Run git pull repeatedly until Git chooses

Q088**EXPERT****In Git, a binary file conflicts. What is the best answer?**

- A. Run git pull repeatedly until Git chooses
- B. Mark the file resolved without reviewing content
- C. choose one version or regenerate it
- D. Delete both conflicted versions

Q089**INTERMEDIATE****In Git, a release tag must be auditable. What is the best answer?**

- A. Move an existing public release tag silently
- B. create a signed annotated tag
- C. Use a lightweight tag and call it cryptographically verified
- D. Store the release version only in a branch name

Q090**ADVANCED****In Git, git tag v1.2 creates a lightweight tag. What is the best answer?**

- A. Use a lightweight tag and call it cryptographically verified
- B. it is only a name pointing at a commit
- C. Store the release version only in a branch name
- D. Move an existing public release tag silently

Q091**ADVANCED****In Git, a tag was created locally but is absent remotely. What is the best answer?**

- A. Use a lightweight tag and call it cryptographically verified
- B. Move an existing public release tag silently
- C. Store the release version only in a branch name
- D. push the tag explicitly

Q092**EXPERT****In Git, a published tag must be moved. What is the best answer?**

- A. Move an existing public release tag silently
- B. Use a lightweight tag and call it cryptographically verified
- C. avoid moving it; create a corrected version tag
- D. Store the release version only in a branch name

Q093

INTERMEDIATE

In Git, a repository contains a submodule. What is the best answer?

- A. Submodule content is embedded in the parent commit
- B. The parent repository always follows the latest submodule branch
- C. Deleting .gitmodules updates every clone automatically
- D. the parent stores a specific submodule commit

Q094

ADVANCED

In Git, a clone shows an empty submodule directory. What is the best answer?

- A. The parent repository always follows the latest submodule branch
- B. Submodule content is embedded in the parent commit
- C. run `git submodule update --init --recursive`
- D. Deleting .gitmodules updates every clone automatically

Q095

ADVANCED

In Git, the submodule advanced but the parent repository did not. What is the best answer?

- A. commit the updated submodule pointer in the parent
- B. The parent repository always follows the latest submodule branch
- C. Submodule content is embedded in the parent commit
- D. Deleting .gitmodules updates every clone automatically

Q096

EXPERT

In Git, a team frequently struggles with submodule coordination. What is the best answer?

- A. Submodule content is embedded in the parent commit
- B. Deleting .gitmodules updates every clone automatically
- C. The parent repository always follows the latest submodule branch
- D. consider whether a package registry or monorepo better matches the workflow

Q097

INTERMEDIATE

In Git, a local branch should track origin/main. What is the best answer?

- A. Delete remote-tracking branches manually after every fetch
- B. Configure all branches to track every remote branch
- C. set its upstream branch
- D. Use `git pull` for a fetch-only operation

Q098

ADVANCED

In Git, git pull unexpectedly creates a merge commit. What is the best answer?

- A. configure or specify rebase/ff-only behavior
- B. Use git pull for a fetch-only operation
- C. Delete remote-tracking branches manually after every fetch
- D. Configure all branches to track every remote branch

Q099

ADVANCED

In Git, you want to update remote references without changing the working tree. What is the best answer?

- A. Configure all branches to track every remote branch
- B. run git fetch
- C. Use git pull for a fetch-only operation
- D. Delete remote-tracking branches manually after every fetch

Q100

EXPERT

In Git, a remote branch was deleted but still appears locally as origin/x. What is the best answer?

- A. fetch with --prune
- B. Configure all branches to track every remote branch
- C. Delete remote-tracking branches manually after every fetch
- D. Use git pull for a fetch-only operation

Q101

INTERMEDIATE

In Git, large binary files make clone size grow rapidly. What is the best answer?

- A. use Git LFS or an artifact repository
- B. Deleting the latest secret commit makes old history safe
- C. Version generated dependencies without a lockfile
- D. Commit changing binaries directly because Git delta compression solves all growth

Q102

ADVANCED

In Git, a secret was committed and then deleted in a later commit. What is the best answer?

- A. rotate it and rewrite history if required
- B. Version generated dependencies without a lockfile
- C. Commit changing binaries directly because Git delta compression solves all growth
- D. Deleting the latest secret commit makes old history safe

Q103**ADVANCED****In Git, git gc is run. What is the best answer?**

- A. Version generated dependencies without a lockfile
- B. it repacks and prunes objects according to retention rules
- C. Deleting the latest secret commit makes old history safe
- D. Commit changing binaries directly because Git delta compression solves all growth

Q104**EXPERT****In Git, a generated dependency directory changes constantly. What is the best answer?**

- A. Version generated dependencies without a lockfile
- B. ignore it and rebuild from a lockfile
- C. Deleting the latest secret commit makes old history safe
- D. Commit changing binaries directly because Git delta compression solves all growth

CI/CD

CI/CD

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CI/CD Questions

Q105**INTERMEDIATE**

In a CI/CD pipeline, a deployment reuses the tag latest and sometimes runs old code. What is the strongest practice?

- A. Reuse latest and delete all Pods before deployment
- B. Trust node image caches to select the intended version
- C. use an immutable build tag or image digest
- D. Rebuild separately in every environment

Q106**ADVANCED**

In a CI/CD pipeline, multiple environments pull the same mutable image tag at different times. What is the strongest practice?

- A. Trust node image caches to select the intended version
- B. promote one immutable artifact across environments
- C. Reuse latest and delete all Pods before deployment
- D. Rebuild separately in every environment

Q107**ADVANCED**

In a CI/CD pipeline, a rollback needs the exact prior container image. What is the strongest practice?

- A. Reuse latest and delete all Pods before deployment
- B. Rebuild separately in every environment
- C. record and redeploy the previous digest
- D. Trust node image caches to select the intended version

Q108**EXPERT**

In a CI/CD pipeline, a pipeline rebuilds source separately for staging and production. What is the strongest practice?

- A. Rebuild separately in every environment
- B. build once, then promote the signed artifact
- C. Reuse latest and delete all Pods before deployment
- D. Trust node image caches to select the intended version

Q109

INTERMEDIATE

In a CI/CD pipeline, ephemeral agents use a long-lived cloud access key. What is the strongest practice?

- A. use workload identity or short-lived federated credentials
- B. Base64-encode a permanent access key
- C. Expose production credentials to all pipeline stages
- D. Store the same long-lived key in more secret stores

Q110

ADVANCED

In a CI/CD pipeline, a secret is stored in a repository variable but printed by set -x. What is the strongest practice?

- A. Base64-encode a permanent access key
- B. Expose production credentials to all pipeline stages
- C. disable command echo and use masked secret handling
- D. Store the same long-lived key in more secret stores

Q111

ADVANCED

In a CI/CD pipeline, a pipeline needs database credentials only during one stage. What is the strongest practice?

- A. Expose production credentials to all pipeline stages
- B. Store the same long-lived key in more secret stores
- C. scope the secret to that job and environment
- D. Base64-encode a permanent access key

Q112

EXPERT

In a CI/CD pipeline, a shared runner can access production credentials. What is the strongest practice?

- A. separate trusted production runners and protected environments
- B. Store the same long-lived key in more secret stores
- C. Base64-encode a permanent access key
- D. Expose production credentials to all pipeline stages

Q113

INTERMEDIATE

In a CI/CD pipeline, unit tests, security scans, and packaging are independent. What is the strongest practice?

- A. Use sleep commands instead of dependency edges
- B. Allow deployment to continue after required tests fail
- C. Run every stage sequentially regardless of dependency
- D. run them in parallel after compilation where dependencies allow

Q114

ADVANCED

In a CI/CD pipeline, deployment starts before test results are complete. What is the strongest practice?

- A. Use sleep commands instead of dependency edges
- B. Allow deployment to continue after required tests fail
- C. make deployment depend on successful quality stages
- D. Run every stage sequentially regardless of dependency

Q115

ADVANCED

In a CI/CD pipeline, a pipeline repeats expensive setup in every job. What is the strongest practice?

- A. Use sleep commands instead of dependency edges
- B. Allow deployment to continue after required tests fail
- C. use a controlled cache keyed by dependency lockfile
- D. Run every stage sequentially regardless of dependency

Q116

EXPERT

In a CI/CD pipeline, a job fails but downstream jobs still run with partial outputs. What is the strongest practice?

- A. Allow deployment to continue after required tests fail
- B. Run every stage sequentially regardless of dependency
- C. Use sleep commands instead of dependency edges
- D. enforce explicit dependencies and fail-fast behavior

Q117

INTERMEDIATE

In a CI/CD pipeline, an artifact must pass from build to deploy. What is the strongest practice?

- A. Use a cache as the authoritative release package
- B. publish it as a versioned pipeline artifact
- C. Rebuild the package during deployment
- D. Transfer artifacts through unversioned shared directories

Q118

ADVANCED

In a CI/CD pipeline, dependency downloads are slow but can be regenerated. What is the strongest practice?

- A. Transfer artifacts through unversioned shared directories
- B. Use a cache as the authoritative release package
- C. Rebuild the package during deployment
- D. use a cache, not a release artifact

Q119

ADVANCED

In a CI/CD pipeline, a cache may contain poisoned content from an untrusted branch. What is the strongest practice?

- A. Transfer artifacts through unversioned shared directories
- B. separate cache keys and trust boundaries
- C. Use a cache as the authoritative release package
- D. Rebuild the package during deployment

Q120

EXPERT

In a CI/CD pipeline, retention deletes old artifacts needed for rollback. What is the strongest practice?

- A. Transfer artifacts through unversioned shared directories
- B. Use a cache as the authoritative release package
- C. align artifact retention with rollback and compliance needs
- D. Rebuild the package during deployment

Q121

INTERMEDIATE

In a CI/CD pipeline, a risky release should reach 5% of users first. What is the strongest practice?

- A. Deploy to all users at once and monitor afterward
- B. Change the database destructively before compatibility is established
- C. use a canary deployment
- D. Use a rolling restart without success criteria

Q122

ADVANCED

In a CI/CD pipeline, two complete environments are available and switching must be instant. What is the strongest practice?

- A. Deploy to all users at once and monitor afterward
- B. Use a rolling restart without success criteria
- C. Change the database destructively before compatibility is established
- D. use blue-green deployment

Q123

ADVANCED

In a CI/CD pipeline, a database migration is not backward compatible. What is the strongest practice?

- A. use an expand-and-contract migration before application cutover
- B. Change the database destructively before compatibility is established
- C. Deploy to all users at once and monitor afterward
- D. Use a rolling restart without success criteria

Q124

EXPERT

In a CI/CD pipeline, a canary has worse latency but equal error rate. What is the strongest practice?

- A. Change the database destructively before compatibility is established
- B. define multi-signal success criteria and stop promotion
- C. Deploy to all users at once and monitor afterward
- D. Use a rolling restart without success criteria

Q125

INTERMEDIATE

In a CI/CD pipeline, a deployment fails health checks immediately. What is the strongest practice?

- A. automatically halt and roll back to the last healthy version
- B. Delete failed resources without restoring the previous version
- C. Continue promotion and investigate after completion
- D. Reverse every schema change automatically regardless of data loss

Q126

ADVANCED

In a CI/CD pipeline, rollback also requires reversing a destructive schema migration. What is the strongest practice?

- A. the release design is unsafe; use forward-compatible migrations
- B. Reverse every schema change automatically regardless of data loss
- C. Continue promotion and investigate after completion
- D. Delete failed resources without restoring the previous version

Q127

ADVANCED

In a CI/CD pipeline, a rollback command exists but has never been tested. What is the strongest practice?

- A. Delete failed resources without restoring the previous version
- B. Reverse every schema change automatically regardless of data loss
- C. Continue promotion and investigate after completion
- D. exercise it regularly in a production-like environment

Q128

EXPERT

In a CI/CD pipeline, a partial regional rollout fails. What is the strongest practice?

- A. Reverse every schema change automatically regardless of data loss
- B. Continue promotion and investigate after completion
- C. Delete failed resources without restoring the previous version
- D. stop further regions and restore the affected region

Q129

INTERMEDIATE

In a CI/CD pipeline, a test fails randomly without code changes. What is the strongest practice?

- A. Run the flaky test only in production
- B. Ignore the test permanently
- C. quarantine only temporarily and fix the nondeterminism
- D. Add unlimited retries and accept the result

Q130

ADVANCED

In a CI/CD pipeline, retries make a flaky test pass. What is the strongest practice?

- A. do not treat retries as the root-cause fix
- B. Ignore the test permanently
- C. Run the flaky test only in production
- D. Add unlimited retries and accept the result

Q131

ADVANCED

In a CI/CD pipeline, tests share a mutable external environment. What is the strongest practice?

- A. isolate test state or provision ephemeral dependencies
- B. Add unlimited retries and accept the result
- C. Run the flaky test only in production
- D. Ignore the test permanently

Q132

EXPERT

In a CI/CD pipeline, a test suite takes 90 minutes. What is the strongest practice?

- A. Add unlimited retries and accept the result
- B. Ignore the test permanently
- C. Run the flaky test only in production
- D. profile, shard, parallelize, and move suitable checks earlier

Q133

INTERMEDIATE

In a CI/CD pipeline, ten independent test groups run sequentially. What is the strongest practice?

- A. Parallelize stateful jobs without isolation
- B. Remove resource limits from shared dependencies
- C. fan them out across parallel jobs
- D. Duplicate deployment jobs across runners

Q134

ADVANCED

In a CI/CD pipeline, parallel jobs overload a shared database. What is the strongest practice?

- A. Duplicate deployment jobs across runners
- B. Remove resource limits from shared dependencies
- C. cap concurrency or isolate test databases
- D. Parallelize stateful jobs without isolation

Q135

ADVANCED

In a CI/CD pipeline, one slow shard determines total duration. What is the strongest practice?

- A. Remove resource limits from shared dependencies
- B. rebalance shards using historical timings
- C. Parallelize stateful jobs without isolation
- D. Duplicate deployment jobs across runners

Q136

EXPERT

In a CI/CD pipeline, parallel jobs create duplicate deployment actions. What is the strongest practice?

- A. Duplicate deployment jobs across runners
- B. Parallelize stateful jobs without isolation
- C. Remove resource limits from shared dependencies
- D. use concurrency locks for non-idempotent environments

Q137

INTERMEDIATE

In a CI/CD pipeline, a secret appears in old build logs. What is the strongest practice?

- A. Use one global cache key for trusted and untrusted branches
- B. Treat release artifacts and caches as interchangeable
- C. revoke it and remove or restrict the logs
- D. Delete rollback artifacts immediately after deployment

Q138

ADVANCED

In a CI/CD pipeline, masking replaces exact secret strings only. What is the strongest practice?

- A. Use one global cache key for trusted and untrusted branches
- B. Treat release artifacts and caches as interchangeable
- C. Delete rollback artifacts immediately after deployment
- D. derived or encoded forms may still leak

Q139

ADVANCED

In a CI/CD pipeline, a pipeline downloads a script and pipes it directly to shell. What is the strongest practice?

- A. pin and verify the downloaded content before execution
- B. Treat release artifacts and caches as interchangeable
- C. Use one global cache key for trusted and untrusted branches
- D. Delete rollback artifacts immediately after deployment

Q140

EXPERT

In a CI/CD pipeline, a pull request from a fork requests privileged secrets. What is the strongest practice?

- A. Use one global cache key for trusted and untrusted branches
- B. Delete rollback artifacts immediately after deployment
- C. do not expose production secrets to untrusted fork jobs
- D. Treat release artifacts and caches as interchangeable

Q141

INTERMEDIATE

In a CI/CD pipeline, coverage falls below the agreed threshold. What is the strongest practice?

- A. Pipe unpinned remote scripts directly into a shell
- B. Rely only on log masking after printing the secret
- C. fail the quality gate or require an approved exception
- D. Expose secrets to forked pull requests

Q142

ADVANCED

In a CI/CD pipeline, a vulnerability scanner finds a critical exploitable issue. What is the strongest practice?

- A. Pipe unpinned remote scripts directly into a shell
- B. block release according to policy
- C. Expose secrets to forked pull requests
- D. Rely only on log masking after printing the secret

Q143

ADVANCED

In a CI/CD pipeline, a static-analysis rule produces many false positives. What is the strongest practice?

- A. Pipe unpinned remote scripts directly into a shell
- B. Expose secrets to forked pull requests
- C. tune the rule with documented ownership rather than disabling all scanning
- D. Rely only on log masking after printing the secret

Q144**EXPERT**

In a CI/CD pipeline, a manual checklist is often skipped. What is the strongest practice?

- A. Pipe unpinned remote scripts directly into a shell
- B. Rely only on log masking after printing the secret
- C. Expose secrets to forked pull requests
- D. automate objective checks in the pipeline

Q145**INTERMEDIATE**

In a CI/CD pipeline, production deployment requires business approval. What is the strongest practice?

- A. Use a manual checklist with no audit trail
- B. use a protected environment approval gate
- C. Disable the scanner when it blocks a release
- D. Convert all failures into warnings

Q146**ADVANCED**

In a CI/CD pipeline, an emergency fix must bypass normal timing. What is the strongest practice?

- A. use a documented break-glass path with audit and follow-up
- B. Disable the scanner when it blocks a release
- C. Convert all failures into warnings
- D. Use a manual checklist with no audit trail

Q147**ADVANCED**

In a CI/CD pipeline, the same person authors and approves a sensitive release. What is the strongest practice?

- A. Convert all failures into warnings
- B. apply separation of duties where risk requires it
- C. Disable the scanner when it blocks a release
- D. Use a manual checklist with no audit trail

Q148**EXPERT**

In a CI/CD pipeline, approval occurs before the final artifact digest exists. What is the strongest practice?

- A. Use a manual checklist with no audit trail
- B. Disable the scanner when it blocks a release
- C. Convert all failures into warnings
- D. move approval to the immutable release candidate

Q149

INTERMEDIATE

In a CI/CD pipeline, you need to know which components are inside an image. What is the strongest practice?

- A. Approve before the final artifact exists
- B. generate an SBOM
- C. Allow any contributor to approve production
- D. Use a shared approver account

Q150

ADVANCED

In a CI/CD pipeline, a registry must verify publisher identity. What is the strongest practice?

- A. Approve before the final artifact exists
- B. Allow any contributor to approve production
- C. sign images and enforce verification at admission
- D. Use a shared approver account

Q151

ADVANCED

In a CI/CD pipeline, a base image receives a critical fix. What is the strongest practice?

- A. Use a shared approver account
- B. Approve before the final artifact exists
- C. rebuild dependent images and verify the new digest
- D. Allow any contributor to approve production

Q152

EXPERT

In a CI/CD pipeline, provenance must show how an artifact was built. What is the strongest practice?

- A. Use a shared approver account
- B. Approve before the final artifact exists
- C. Allow any contributor to approve production
- D. produce attestations from a trusted build system

Q153

INTERMEDIATE

In a CI/CD pipeline, deployment succeeds but users see errors. What is the strongest practice?

- A. Trust unsigned artifacts if the filename looks correct
- B. Assume old images inherit base-image patches automatically
- C. Accept provenance from an untrusted build worker
- D. add post-deployment synthetic and service-level verification

Q154

ADVANCED

In a CI/CD pipeline, a smoke test checks only HTTP 200 on /health. What is the strongest practice?

- A. Trust unsigned artifacts if the filename looks correct
- B. Accept provenance from an untrusted build worker
- C. add critical user-journey checks
- D. Assume old images inherit base-image patches automatically

Q155

ADVANCED

In a CI/CD pipeline, metrics are delayed by ten minutes. What is the strongest practice?

- A. Accept provenance from an untrusted build worker
- B. Trust unsigned artifacts if the filename looks correct
- C. Assume old images inherit base-image patches automatically
- D. do not make an instant automated decision from stale data alone

Q156

EXPERT

In a CI/CD pipeline, a deployment changes no desired-state manifest because the tag is unchanged. What is the strongest practice?

- A. Accept provenance from an untrusted build worker
- B. ensure the deployment specification changes to an immutable version
- C. Assume old images inherit base-image patches automatically
- D. Trust unsigned artifacts if the filename looks correct

Docker

DOCKER

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Docker Questions

Q157**INTERMEDIATE**

In Docker, a port is published as 127.0.0.1:8080:8080. Which statement or action is correct?

- A. only the Docker host can reach it through loopback
- B. EXPOSE automatically publishes the port to remote hosts
- C. A host mapping works even when the app binds only to container loopback
- D. --network host is required for every externally reachable service

Q158**ADVANCED**

In Docker, remote clients must reach container port 8080. Which statement or action is correct?

- A. publish it on an appropriate external host interface
- B. EXPOSE automatically publishes the port to remote hosts
- C. --network host is required for every externally reachable service
- D. A host mapping works even when the app binds only to container loopback

Q159**ADVANCED**

In Docker, the app listens on 127.0.0.1 inside the container. Which statement or action is correct?

- A. A host mapping works even when the app binds only to container loopback
- B. --network host is required for every externally reachable service
- C. publish mapping alone will not make it reachable
- D. EXPOSE automatically publishes the port to remote hosts

Q160**EXPERT**

In Docker, a port is exposed in the Dockerfile but not published. Which statement or action is correct?

- A. EXPOSE documents intent but does not create a host mapping
- B. A host mapping works even when the app binds only to container loopback
- C. EXPOSE automatically publishes the port to remote hosts
- D. --network host is required for every externally reachable service

Q161**INTERMEDIATE**

In Docker, state is written only to the container writable layer. Which statement or action is correct?

- A. Commit the running container after every write
- B. Increase the writable layer and treat it as a backup
- C. it is lost when the container is removed
- D. Use restart: always as persistent storage

Q162

ADVANCED

In Docker, database data must survive container recreation. Which statement or action is correct?

- A. Commit the running container after every write
- B. Use restart: always as persistent storage
- C. Increase the writable layer and treat it as a backup
- D. mount a named volume or managed persistent storage

Q163

ADVANCED

In Docker, a bind mount points to a host path. Which statement or action is correct?

- A. Commit the running container after every write
- B. Use restart: always as persistent storage
- C. host filesystem permissions and path existence matter
- D. Increase the writable layer and treat it as a backup

Q164

EXPERT

In Docker, docker compose down -v is run. Which statement or action is correct?

- A. named volumes declared by the project are removed
- B. Commit the running container after every write
- C. Use restart: always as persistent storage
- D. Increase the writable layer and treat it as a backup

Q165

INTERMEDIATE

In Docker, a Go application image includes the full compiler toolchain. Which statement or action is correct?

- A. Pass secrets through ARG and remove them later
- B. Install compilers in the final runtime image
- C. use a multi-stage build
- D. Copy the entire build context into every stage

Q166

ADVANCED

In Docker, the final image needs one binary and CA certificates. Which statement or action is correct?

- A. copy only required runtime assets into a minimal stage
- B. Pass secrets through ARG and remove them later
- C. Install compilers in the final runtime image
- D. Copy the entire build context into every stage

Q167

ADVANCED

In Docker, a build secret is passed with ARG and remains in history. Which statement or action is correct?

- A. Pass secrets through ARG and remove them later
- B. Copy the entire build context into every stage
- C. use BuildKit secret mounts
- D. Install compilers in the final runtime image

Q168

EXPERT

In Docker, dependency download changes rarely. Which statement or action is correct?

- A. copy lockfiles and install dependencies before copying changing source
- B. Pass secrets through ARG and remove them later
- C. Install compilers in the final runtime image
- D. Copy the entire build context into every stage

Q169

INTERMEDIATE

In Docker, a source file changes and every later Dockerfile step rebuilds. Which statement or action is correct?

- A. Split package update and install into unrelated layers
- B. Docker invalidates that layer and all following layers
- C. Put frequently changing source before dependency installation
- D. Include logs and local caches in the build context

Q170

ADVANCED

In Docker, apt-get update and install are split across layers. Which statement or action is correct?

- A. Split package update and install into unrelated layers
- B. Include logs and local caches in the build context
- C. combine them and clean package lists in one RUN
- D. Put frequently changing source before dependency installation

Q171

ADVANCED

In Docker, a cache is reused across incompatible architectures. Which statement or action is correct?

- A. Include logs and local caches in the build context
- B. Put frequently changing source before dependency installation
- C. Split package update and install into unrelated layers
- D. use architecture-aware cache keys

Q172

EXPERT

In Docker, the build context includes gigabytes of logs. Which statement or action is correct?

- A. use .dockerignore
- B. Split package update and install into unrelated layers
- C. Put frequently changing source before dependency installation
- D. Include logs and local caches in the build context

Q173

INTERMEDIATE

In Docker, a container process is running but the app is unusable. Which statement or action is correct?

- A. Assume a running process proves application health
- B. Expect standalone Docker to replace unhealthy containers automatically
- C. add a meaningful HEALTHCHECK
- D. Use a dependency-heavy probe that can trigger cascading failure

Q174

ADVANCED

In Docker, a health check starts before a slow app initializes. Which statement or action is correct?

- A. Assume a running process proves application health
- B. Expect standalone Docker to replace unhealthy containers automatically
- C. configure start-period or orchestrator startup checks
- D. Use a dependency-heavy probe that can trigger cascading failure

Q175

ADVANCED

In Docker, a health check calls an expensive dependency on every probe. Which statement or action is correct?

- A. Assume a running process proves application health
- B. use a lightweight local check
- C. Expect standalone Docker to replace unhealthy containers automatically
- D. Use a dependency-heavy probe that can trigger cascading failure

Q176

EXPERT

In Docker, Docker marks a container unhealthy. Which statement or action is correct?

- A. Assume a running process proves application health
- B. it records health status but standalone Docker does not automatically replace it
- C. Use a dependency-heavy probe that can trigger cascading failure
- D. Expect standalone Docker to replace unhealthy containers automatically

Q177

INTERMEDIATE

In Docker, a shell script is PID 1 and does not forward SIGTERM. Which statement or action is correct?

- A. Ignore child-process reaping in containers
- B. use exec to launch the application
- C. Keep a shell wrapper as PID 1 without signal forwarding
- D. Use SIGKILL as the normal shutdown method

Q178

ADVANCED

In Docker, PID 1 does not reap orphaned children. Which statement or action is correct?

- A. Use SIGKILL as the normal shutdown method
- B. Ignore child-process reaping in containers
- C. Keep a shell wrapper as PID 1 without signal forwarding
- D. use an init process such as tini or correct the application

Q179

ADVANCED

In Docker, docker stop exceeds the grace period. Which statement or action is correct?

- A. Use SIGKILL as the normal shutdown method
- B. Docker sends SIGKILL after the timeout
- C. Keep a shell wrapper as PID 1 without signal forwarding
- D. Ignore child-process reaping in containers

Q180

EXPERT

In Docker, an app needs time to drain connections. Which statement or action is correct?

- A. Keep a shell wrapper as PID 1 without signal forwarding
- B. Ignore child-process reaping in containers
- C. Use SIGKILL as the normal shutdown method
- D. implement graceful shutdown and set a suitable stop timeout

Q181

INTERMEDIATE

In Docker, containers run as root by default. Which statement or action is correct?

- A. Mount the Docker socket because it is read-only by default
- B. Run every container as root for compatibility
- C. Use --privileged instead of a narrow capability
- D. set a non-root USER where possible

Q182

ADVANCED

In Docker, the app only needs to bind a low port. Which statement or action is correct?

- A. Use --privileged instead of a narrow capability
- B. use a targeted capability or higher unprivileged port
- C. Mount the Docker socket because it is read-only by default
- D. Run every container as root for compatibility

Q183

ADVANCED

In Docker, the Docker socket is mounted into a container. Which statement or action is correct?

- A. Run every container as root for compatibility
- B. that container effectively has host-level control
- C. Mount the Docker socket because it is read-only by default
- D. Use --privileged instead of a narrow capability

Q184

EXPERT

In Docker, rootless Docker is used. Which statement or action is correct?

- A. Use --privileged instead of a narrow capability
- B. Run every container as root for compatibility
- C. daemon and containers operate without root privileges on the host
- D. Mount the Docker socket because it is read-only by default

Q185

INTERMEDIATE

In Docker, an image tag is changed to point at new content. Which statement or action is correct?

- A. Assume one tag can represent only one architecture image
- B. Treat tags as immutable content identifiers
- C. the digest distinguishes the exact image
- D. Use the image name without a tag for reproducibility

Q186

ADVANCED

In Docker, two registries mirror the same digest. Which statement or action is correct?

- A. Assume one tag can represent only one architecture image
- B. the image content is identical if the digest algorithm and manifest match
- C. Treat tags as immutable content identifiers
- D. Use the image name without a tag for reproducibility

Q187

ADVANCED

In Docker, a deployment must be reproducible. Which statement or action is correct?

- A. Use the image name without a tag for reproducibility
- B. Assume one tag can represent only one architecture image
- C. Treat tags as immutable content identifiers
- D. pin the image digest

Q188

EXPERT

In Docker, a multi-architecture tag is inspected. Which statement or action is correct?

- A. Treat tags as immutable content identifiers
- B. it may reference a manifest list with platform-specific images
- C. Use the image name without a tag for reproducibility
- D. Assume one tag can represent only one architecture image

Q189

INTERMEDIATE

In Docker, a file is deleted in a later image layer. Which statement or action is correct?

- A. bytes from the earlier layer may still remain in the image
- B. Store databases in the copy-on-write layer for best durability
- C. Deleting a file in a later layer removes its bytes from all prior layers
- D. Create and delete temporary files in separate layers

Q190

ADVANCED

In Docker, many temporary build files are created and deleted in separate RUN steps. Which statement or action is correct?

- A. Create and delete temporary files in separate layers
- B. Store databases in the copy-on-write layer for best durability
- C. the image can remain unnecessarily large
- D. Deleting a file in a later layer removes its bytes from all prior layers

Q191

ADVANCED

In Docker, copy-on-write storage receives heavy database writes. Which statement or action is correct?

- A. Deleting a file in a later layer removes its bytes from all prior layers
- B. Store databases in the copy-on-write layer for best durability
- C. Create and delete temporary files in separate layers
- D. a volume is generally more suitable

Q192**EXPERT**

In Docker, docker system prune is run. Which statement or action is correct?

- A. unused objects can be deleted
- B. Store databases in the copy-on-write layer for best durability
- C. Create and delete temporary files in separate layers
- D. Deleting a file in a later layer removes its bytes from all prior layers

Q193**INTERMEDIATE**

In Docker, a container exceeds its CPU quota. Which statement or action is correct?

- A. the kernel throttles CPU time
- B. Memory is throttled exactly like CPU
- C. No limits means the container cannot affect the host
- D. Requests automatically resize the container

Q194**ADVANCED**

In Docker, a container exceeds its memory limit. Which statement or action is correct?

- A. it may be OOM-killed
- B. No limits means the container cannot affect the host
- C. Requests automatically resize the container
- D. Memory is throttled exactly like CPU

Q195**ADVANCED**

In Docker, no memory limit is set. Which statement or action is correct?

- A. Requests automatically resize the container
- B. Memory is throttled exactly like CPU
- C. No limits means the container cannot affect the host
- D. the container can compete for host memory

Q196**EXPERT**

In Docker, a JVM ignores container limits on an old runtime. Which statement or action is correct?

- A. Requests automatically resize the container
- B. Memory is throttled exactly like CPU
- C. upgrade or explicitly configure container-aware memory settings
- D. No limits means the container cannot affect the host

Q197

INTERMEDIATE

In Docker, two containers on the same user-defined bridge use service names. Which statement or action is correct?

- A. Rely on the default bridge for complete service discovery
- B. Place all workloads on one flat network
- C. Docker provides embedded DNS for that network
- D. Use localhost to reach another container

Q198

ADVANCED

In Docker, a container uses localhost to reach another container. Which statement or action is correct?

- A. localhost refers to itself
- B. Place all workloads on one flat network
- C. Use localhost to reach another container
- D. Rely on the default bridge for complete service discovery

Q199

ADVANCED

In Docker, a container must join two isolated networks. Which statement or action is correct?

- A. Place all workloads on one flat network
- B. connect it to both networks deliberately
- C. Use localhost to reach another container
- D. Rely on the default bridge for complete service discovery

Q200

EXPERT

In Docker, the default bridge requires legacy links for names. Which statement or action is correct?

- A. Rely on the default bridge for complete service discovery
- B. prefer a user-defined bridge network
- C. Use localhost to reach another container
- D. Place all workloads on one flat network

Q201

INTERMEDIATE

In Docker, Compose depends_on is used. Which statement or action is correct?

- A. Bake runtime secrets into the image
- B. depends_on guarantees the application is ready
- C. it controls start order but not application readiness unless health conditions are configured
- D. Startup ordering removes the need for retry logic

Q202

ADVANCED

In Docker, an app should wait for a database health check. Which statement or action is correct?

- A. use a healthcheck plus a readiness-aware dependency or retry logic
- B. depends_on guarantees the application is ready
- C. Bake runtime secrets into the image
- D. Startup ordering removes the need for retry logic

Q203

ADVANCED

In Docker, a dependency restarts later. Which statement or action is correct?

- A. the application still needs reconnect logic
- B. Startup ordering removes the need for retry logic
- C. Bake runtime secrets into the image
- D. depends_on guarantees the application is ready

Q204

EXPERT

In Docker, Compose secrets are available. Which statement or action is correct?

- A. Bake runtime secrets into the image
- B. mount them as files rather than baking them into images
- C. Startup ordering removes the need for retry logic
- D. depends_on guarantees the application is ready

Q205

INTERMEDIATE

In Docker, json-file logs grow without bound. Which statement or action is correct?

- A. Log credentials and depend on transport encryption
- B. Disable log rotation to preserve every event
- C. configure log rotation options or a logging driver
- D. Write logs only inside the container filesystem

Q206

ADVANCED

In Docker, logs are written only inside the container filesystem. Which statement or action is correct?

- A. emit to stdout/stderr for centralized collection
- B. Log credentials and depend on transport encryption
- C. Disable log rotation to preserve every event
- D. Write logs only inside the container filesystem

Q207**ADVANCED**

In Docker, sensitive data appears in application logs. Which statement or action is correct?

- A. Log credentials and depend on transport encryption
- B. Write logs only inside the container filesystem
- C. Disable log rotation to preserve every event
- D. remove or redact it at the source

Q208**EXPERT**

In Docker, the logging backend is unavailable. Which statement or action is correct?

- A. Log credentials and depend on transport encryption
- B. Disable log rotation to preserve every event
- C. Write logs only inside the container filesystem
- D. define buffering and backpressure behavior carefully

Kubernetes

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Kubernetes Questions

Q209**INTERMEDIATE**

In Kubernetes, a Service has ENDPOINTS while Pods are Running. What is the best answer?

- A. check whether the Service selector matches Pod labels
- B. Convert the Service to NodePort
- C. Change the application to listen on the Service port
- D. Restart CoreDNS before checking selectors

Q210**ADVANCED**

In Kubernetes, EndpointSlices list no addresses. What is the best answer?

- A. Convert the Service to NodePort
- B. verify labels and Pod readiness conditions
- C. Change the application to listen on the Service port
- D. Restart CoreDNS before checking selectors

Q211**ADVANCED**

In Kubernetes, the Service port is 80 and targetPort is 8080. What is the best answer?

- A. Change the application to listen on the Service port
- B. Convert the Service to NodePort
- C. Restart CoreDNS before checking selectors
- D. the application can still listen on 8080

Q212**EXPERT**

In Kubernetes, a headless Service uses clusterIP: None. What is the best answer?

- A. Restart CoreDNS before checking selectors
- B. DNS returns Pod endpoint addresses instead of one virtual IP
- C. Change the application to listen on the Service port
- D. Convert the Service to NodePort

Q213**INTERMEDIATE**

In Kubernetes, an app needs 90 seconds to initialize but liveness starts at 10 seconds. What is the best answer?

- A. add a startupProbe with enough allowance
- B. Increase replica count to hide startup failures
- C. Replace liveness with readiness only
- D. Remove all probes permanently

Q214

ADVANCED

In Kubernetes, readiness fails while liveness succeeds. What is the best answer?

- A. Remove all probes permanently
- B. the Pod stays running but is removed from Service traffic
- C. Increase replica count to hide startup failures
- D. Replace liveness with readiness only

Q215

ADVANCED

In Kubernetes, liveness checks a downstream database. What is the best answer?

- A. Remove all probes permanently
- B. Increase replica count to hide startup failures
- C. this can cause cascading restarts
- D. Replace liveness with readiness only

Q216

EXPERT

In Kubernetes, a probe timeout is shorter than normal response latency. What is the best answer?

- A. tune the probe and fix latency rather than accepting false failures
- B. Increase replica count to hide startup failures
- C. Remove all probes permanently
- D. Replace liveness with readiness only

Q217

INTERMEDIATE

In Kubernetes, maxUnavailable is 0, maxSurge is 1, and the cluster has no spare capacity. What is the best answer?

- A. An old Pod is always deleted first regardless of maxUnavailable
- B. The controller automatically adds nodes
- C. the rollout stalls waiting for a surge Pod
- D. The scheduler ignores CPU requests during rollout

Q218

ADVANCED

In Kubernetes, maxUnavailable is 1 and no surge capacity exists. What is the best answer?

- A. The scheduler ignores CPU requests during rollout
- B. one old Pod may be removed to free capacity
- C. An old Pod is always deleted first regardless of maxUnavailable
- D. The controller automatically adds nodes

Q219

ADVANCED

In Kubernetes, a Deployment rollout is paused. What is the best answer?

- A. The controller automatically adds nodes
- B. new changes are recorded but not progressed until resumed
- C. An old Pod is always deleted first regardless of maxUnavailable
- D. The scheduler ignores CPU requests during rollout

Q220

EXPERT

In Kubernetes, progressDeadlineSeconds is exceeded. What is the best answer?

- A. The controller automatically adds nodes
- B. The scheduler ignores CPU requests during rollout
- C. the Deployment reports ProgressDeadlineExceeded
- D. An old Pod is always deleted first regardless of maxUnavailable

Q221

INTERMEDIATE

In Kubernetes, a container uses 1.5 CPU with a 1 CPU limit. What is the best answer?

- A. it is throttled
- B. The scheduler automatically raises limits
- C. The Pod is rescheduled before any limit is enforced
- D. Both CPU and memory are only throttled

Q222

ADVANCED

In Kubernetes, a container uses 2.5 GiB with a 2 GiB memory limit. What is the best answer?

- A. The Pod is rescheduled before any limit is enforced
- B. it may be OOMKilled
- C. Both CPU and memory are only throttled
- D. The scheduler automatically raises limits

Q223

ADVANCED

In Kubernetes, a Pod request is 500m CPU. What is the best answer?

- A. The Pod is rescheduled before any limit is enforced
- B. The scheduler automatically raises limits
- C. the scheduler reserves 0.5 CPU for placement calculations
- D. Both CPU and memory are only throttled

Q224

EXPERT

In Kubernetes, requests equal limits for all containers. What is the best answer?

- A. Both CPU and memory are only throttled
- B. the Pod can receive Guaranteed QoS
- C. The scheduler automatically raises limits
- D. The Pod is rescheduled before any limit is enforced

Q225

INTERMEDIATE

In Kubernetes, nodes fail due to affinity and untolerated taints. What is the best answer?

- A. Convert required affinity into a Service selector
- B. Increase priority to override all taints and affinity
- C. adding priority alone cannot make incompatible nodes eligible
- D. Restart the scheduler

Q226

ADVANCED

In Kubernetes, a Pod has a NoSchedule taint mismatch. What is the best answer?

- A. add an appropriate toleration or use an untainted node
- B. Restart the scheduler
- C. Convert required affinity into a Service selector
- D. Increase priority to override all taints and affinity

Q227

ADVANCED

In Kubernetes, requiredDuringScheduling affinity has no matching node. What is the best answer?

- A. Restart the scheduler
- B. the Pod remains Pending
- C. Increase priority to override all taints and affinity
- D. Convert required affinity into a Service selector

Q228

EXPERT

In Kubernetes, preferredDuringScheduling affinity cannot be satisfied. What is the best answer?

- A. Increase priority to override all taints and affinity
- B. Restart the scheduler
- C. Convert required affinity into a Service selector
- D. the scheduler may still place the Pod elsewhere

Q229

INTERMEDIATE

In Kubernetes, a PVC remains Pending. What is the best answer?

- A. A PVC becomes bound by converting the Service to LoadBalancer
- B. ReadWriteOnce means exactly one Pod in all cases
- C. Delete the StorageClass without checking topology
- D. check StorageClass, provisioner, access mode, and topology

Q230

ADVANCED

In Kubernetes, a volume uses ReadWriteOnce. What is the best answer?

- A. ReadWriteOnce means exactly one Pod in all cases
- B. A PVC becomes bound by converting the Service to LoadBalancer
- C. it is normally writable by one node at a time, not necessarily one Pod
- D. Delete the StorageClass without checking topology

Q231

ADVANCED

In Kubernetes, a StatefulSet uses volumeClaimTemplates. What is the best answer?

- A. Delete the StorageClass without checking topology
- B. each replica receives a stable PVC
- C. A PVC becomes bound by converting the Service to LoadBalancer
- D. ReadWriteOnce means exactly one Pod in all cases

Q232

EXPERT

In Kubernetes, a zonal PV is bound before Pod scheduling in the wrong zone. What is the best answer?

- A. ReadWriteOnce means exactly one Pod in all cases
- B. Delete the StorageClass without checking topology
- C. WaitForFirstConsumer can align storage provisioning with Pod placement
- D. A PVC becomes bound by converting the Service to LoadBalancer

Q233

INTERMEDIATE

In Kubernetes, a Pod is in CrashLoopBackOff. What is the best answer?

- A. inspect previous container logs and termination reason
- B. Increase the Service timeout to stop restarts
- C. Delete the Pod repeatedly without reading previous logs
- D. CrashLoopBackOff is the root application error

Q234

ADVANCED

In Kubernetes, the container exits with code 0 repeatedly under a Deployment. What is the best answer?

- A. CrashLoopBackOff is the root application error
- B. Increase the Service timeout to stop restarts
- C. Delete the Pod repeatedly without reading previous logs
- D. the main process is completing when a long-running service is expected

Q235

ADVANCED

In Kubernetes, OOMKilled appears in lastState. What is the best answer?

- A. Increase the Service timeout to stop restarts
- B. review memory usage and limits
- C. Delete the Pod repeatedly without reading previous logs
- D. CrashLoopBackOff is the root application error

Q236

EXPERT

In Kubernetes, the backoff delay increases. What is the best answer?

- A. CrashLoopBackOff is the root application error
- B. Delete the Pod repeatedly without reading previous logs
- C. Increase the Service timeout to stop restarts
- D. Kubernetes is spacing repeated restart attempts

Q237

INTERMEDIATE

In Kubernetes, ImagePullBackOff occurs. What is the best answer?

- A. Use a mutable tag and imagePullPolicy Never
- B. Create the pull Secret in any namespace
- C. inspect image name, tag, registry reachability, and pull credentials
- D. Restart the API server

Q238

ADVANCED

In Kubernetes, a private registry Secret exists in another namespace. What is the best answer?

- A. Use a mutable tag and imagePullPolicy Never
- B. it cannot be referenced directly by the Pod
- C. Restart the API server
- D. Create the pull Secret in any namespace

Q239

ADVANCED

In Kubernetes, imagePullPolicy is IfNotPresent and a mutable tag is reused. What is the best answer?

- A. Restart the API server
- B. a cached old image may run
- C. Use a mutable tag and imagePullPolicy Never
- D. Create the pull Secret in any namespace

Q240

EXPERT

In Kubernetes, the image supports only amd64 but the node is arm64. What is the best answer?

- A. Create the pull Secret in any namespace
- B. Restart the API server
- C. the pull or start may fail due to platform mismatch
- D. Use a mutable tag and imagePullPolicy Never

Q241

INTERMEDIATE

In Kubernetes, ClusterIP works but service DNS fails. What is the best answer?

- A. Increase the backend CPU limit
- B. Restart the application before testing cluster DNS
- C. check CoreDNS Pods, logs, ConfigMap, and kube-dns endpoints
- D. Convert the backend Service to NodePort

Q242

ADVANCED

In Kubernetes, a short name resolves in one namespace but not another. What is the best answer?

- A. Restart the application before testing cluster DNS
- B. DNS search paths are namespace-aware
- C. Increase the backend CPU limit
- D. Convert the backend Service to NodePort

Q243

ADVANCED

In Kubernetes, a Pod uses dnsPolicy: Default. What is the best answer?

- A. Restart the application before testing cluster DNS
- B. Increase the backend CPU limit
- C. Convert the backend Service to NodePort
- D. it inherits node resolver settings rather than cluster DNS

Q244**EXPERT**

In Kubernetes, ndots is high and external lookups are slow. What is the best answer?

- A. Convert the backend Service to NodePort
- B. Increase the backend CPU limit
- C. Restart the application before testing cluster DNS
- D. search-domain attempts may add latency

Q245**INTERMEDIATE**

In Kubernetes, a PDB sets minAvailable to all replicas. What is the best answer?

- A. A PDB blocks every replica-count change
- B. voluntary node drains may be blocked
- C. Set minAvailable above the replica count
- D. A PDB prevents node hardware failure

Q246**ADVANCED**

In Kubernetes, a node crashes unexpectedly. What is the best answer?

- A. Set minAvailable above the replica count
- B. a PDB cannot prevent involuntary disruption
- C. A PDB blocks every replica-count change
- D. A PDB prevents node hardware failure

Q247**ADVANCED**

In Kubernetes, a single-replica app has minAvailable: 1. What is the best answer?

- A. A PDB prevents node hardware failure
- B. Set minAvailable above the replica count
- C. A PDB blocks every replica-count change
- D. maintenance eviction cannot proceed without violating the PDB

Q248**EXPERT**

In Kubernetes, a Deployment scales replicas down. What is the best answer?

- A. Set minAvailable above the replica count
- B. A PDB blocks every replica-count change
- C. the controller's intentional scale is not blocked by a PDB
- D. A PDB prevents node hardware failure

Q249

INTERMEDIATE

In Kubernetes, HPA uses CPU utilization but Pods lack CPU requests. What is the best answer?

- A. Let multiple controllers write the same replica field
- B. Use only memory for every garbage-collected runtime
- C. HPA works without resource requests in all cases
- D. CPU utilization scaling may not calculate correctly

Q250

ADVANCED

In Kubernetes, HPA and a manual operator both change replicas. What is the best answer?

- A. Use only memory for every garbage-collected runtime
- B. their controllers can fight each other
- C. HPA works without resource requests in all cases
- D. Let multiple controllers write the same replica field

Q251

ADVANCED

In Kubernetes, traffic spikes faster than new Pods become ready. What is the best answer?

- A. HPA works without resource requests in all cases
- B. use suitable stabilization, capacity headroom, and possibly queue metrics
- C. Let multiple controllers write the same replica field
- D. Use only memory for every garbage-collected runtime

Q252

EXPERT

In Kubernetes, memory usage never falls after load ends. What is the best answer?

- A. memory-based HPA may not scale down predictably
- B. HPA works without resource requests in all cases
- C. Use only memory for every garbage-collected runtime
- D. Let multiple controllers write the same replica field

Q253

INTERMEDIATE

In Kubernetes, a ServiceAccount receives Forbidden from the API. What is the best answer?

- A. RoleBindings can grant permissions in all namespaces
- B. Grant cluster-admin to solve Forbidden errors
- C. check Role/ClusterRole bindings and namespace scope
- D. A ClusterRoleBinding is required for every namespaced permission

Q254

ADVANCED

In Kubernetes, a RoleBinding references a ClusterRole. What is the best answer?

- A. RoleBindings can grant permissions in all namespaces
- B. Grant cluster-admin to solve Forbidden errors
- C. the permissions apply only in the RoleBinding namespace
- D. A ClusterRoleBinding is required for every namespaced permission

Q255

ADVANCED

In Kubernetes, an app only needs to list ConfigMaps in one namespace. What is the best answer?

- A. grant a namespaced Role with list on configmaps
- B. RoleBindings can grant permissions in all namespaces
- C. A ClusterRoleBinding is required for every namespaced permission
- D. Grant cluster-admin to solve Forbidden errors

Q256

EXPERT

In Kubernetes, kubectl auth can-i is used with --as. What is the best answer?

- A. RoleBindings can grant permissions in all namespaces
- B. it tests effective authorization for the impersonated identity
- C. A ClusterRoleBinding is required for every namespaced permission
- D. Grant cluster-admin to solve Forbidden errors

Q257

INTERMEDIATE

In Kubernetes, Ingress returns the controller's default 404. What is the best answer?

- A. check host/path rules and ingressClass selection
- B. A 504 means the controller found no matching Ingress rule
- C. A default 404 proves the application is down
- D. A 502 is always a DNS error

Q258

ADVANCED

In Kubernetes, Ingress returns 502 and endpoints exist. What is the best answer?

- A. A 504 means the controller found no matching Ingress rule
- B. A 502 is always a DNS error
- C. check backend port, protocol, readiness, and application reachability
- D. A default 404 proves the application is down

Q259**ADVANCED**

In Kubernetes, Ingress returns 504. What is the best answer?

- A. A 504 means the controller found no matching Ingress rule
- B. the upstream connection or response timed out
- C. A default 404 proves the application is down
- D. A 502 is always a DNS error

Q260**EXPERT**

In Kubernetes, two controllers watch the same Ingress without clear classing. What is the best answer?

- A. A default 404 proves the application is down
- B. A 502 is always a DNS error
- C. A 504 means the controller found no matching Ingress rule
- D. routing can be inconsistent or duplicated

Cloud

CLOUD

Questions 261-312 | 52 questions

Cloud Questions

Q261**INTERMEDIATE**

In cloud architecture, an application spans three AZs but its database is one VM in one AZ. Which conclusion or design is best?

- A. Nightly backups provide automatic high availability
- B. Placing all tiers in one zone improves resilience
- C. A load balancer removes every stateful single point of failure
- D. the database is a single point of failure

Q262**ADVANCED**

In cloud architecture, nightly backups exist for a single database VM. Which conclusion or design is best?

- A. Nightly backups provide automatic high availability
- B. A load balancer removes every stateful single point of failure
- C. Placing all tiers in one zone improves resilience
- D. backups improve recovery but not high availability

Q263**ADVANCED**

In cloud architecture, a managed database offers synchronous standby failover. Which conclusion or design is best?

- A. Nightly backups provide automatic high availability
- B. A load balancer removes every stateful single point of failure
- C. Placing all tiers in one zone improves resilience
- D. it reduces infrastructure failure downtime

Q264**EXPERT**

In cloud architecture, all tiers use one regional control plane. Which conclusion or design is best?

- A. Nightly backups provide automatic high availability
- B. a regional failure still requires a cross-region strategy
- C. Placing all tiers in one zone improves resilience
- D. A load balancer removes every stateful single point of failure

Q265

INTERMEDIATE

In cloud architecture, instances are distributed across availability zones. Which conclusion or design is best?

- A. Zonal state automatically moves with compute
- B. Cross-zone traffic has no cost or latency tradeoff
- C. A multi-AZ label guarantees balanced healthy capacity
- D. the design tolerates a single-zone compute failure if dependencies also do

Q266

ADVANCED

In cloud architecture, a load balancer has healthy targets in only one AZ. Which conclusion or design is best?

- A. Cross-zone traffic has no cost or latency tradeoff
- B. A multi-AZ label guarantees balanced healthy capacity
- C. Zonal state automatically moves with compute
- D. the service still has a zonal concentration risk

Q267

ADVANCED

In cloud architecture, stateful storage is zonal. Which conclusion or design is best?

- A. failover may require replicated or re-created storage in another zone
- B. A multi-AZ label guarantees balanced healthy capacity
- C. Zonal state automatically moves with compute
- D. Cross-zone traffic has no cost or latency tradeoff

Q268

EXPERT

In cloud architecture, cross-zone traffic has cost and latency implications. Which conclusion or design is best?

- A. Zonal state automatically moves with compute
- B. measure them while preserving failure isolation
- C. Cross-zone traffic has no cost or latency tradeoff
- D. A multi-AZ label guarantees balanced healthy capacity

Q269

INTERMEDIATE

In cloud architecture, CPU-based autoscaling reacts after latency has already spiked. Which conclusion or design is best?

- A. add predictive or demand-based signals and capacity headroom
- B. Autoscaling reacts instantly with no startup delay
- C. Scale workers without considering downstream limits
- D. Terminate instances immediately during scale-in

Q270

ADVANCED

In cloud architecture, scale-out launches take ten minutes. Which conclusion or design is best?

- A. Scale workers without considering downstream limits
- B. Terminate instances immediately during scale-in
- C. maintain warm capacity or faster startup paths
- D. Autoscaling reacts instantly with no startup delay

Q271

ADVANCED

In cloud architecture, a downstream database cannot handle unlimited workers. Which conclusion or design is best?

- A. cap scaling and apply backpressure
- B. Scale workers without considering downstream limits
- C. Terminate instances immediately during scale-in
- D. Autoscaling reacts instantly with no startup delay

Q272

EXPERT

In cloud architecture, scale-in terminates busy instances abruptly. Which conclusion or design is best?

- A. Autoscaling reacts instantly with no startup delay
- B. Scale workers without considering downstream limits
- C. Terminate instances immediately during scale-in
- D. use connection draining and scale-in protection where needed

Q273

INTERMEDIATE

In cloud architecture, object storage advertises very high durability. Which conclusion or design is best?

- A. Storage capacity is the only relevant object-storage cost
- B. Durability and availability are the same metric
- C. Versioning prevents all public-access mistakes
- D. it protects stored object integrity, not necessarily application availability

Q274

ADVANCED

In cloud architecture, versioning is enabled. Which conclusion or design is best?

- A. Durability and availability are the same metric
- B. accidental overwrites and deletes can be recovered more easily
- C. Storage capacity is the only relevant object-storage cost
- D. Versioning prevents all public-access mistakes

Q275

ADVANCED

In cloud architecture, a public bucket policy is added accidentally. Which conclusion or design is best?

- A. Storage capacity is the only relevant object-storage cost
- B. Durability and availability are the same metric
- C. prevent it with organization controls and continuous policy checks
- D. Versioning prevents all public-access mistakes

Q276

EXPERT

In cloud architecture, millions of tiny objects create cost overhead. Which conclusion or design is best?

- A. Storage capacity is the only relevant object-storage cost
- B. Versioning prevents all public-access mistakes
- C. Durability and availability are the same metric
- D. request and metadata costs may dominate storage cost

Q277

INTERMEDIATE

In cloud architecture, global users download the same static content. Which conclusion or design is best?

- A. Use identical TTL expiry for all hot keys
- B. A CDN removes the need to secure private content
- C. Cache personalized responses under one global key
- D. a CDN reduces latency and origin load

Q278

ADVANCED

In cloud architecture, personalized responses are cached without a correct cache key. Which conclusion or design is best?

- A. A CDN removes the need to secure private content
- B. Use identical TTL expiry for all hot keys
- C. Cache personalized responses under one global key
- D. users may receive another user's content

Q279

ADVANCED

In cloud architecture, an origin is overloaded during cache expiry. Which conclusion or design is best?

- A. A CDN removes the need to secure private content
- B. use staggered TTLs, request coalescing, or stale serving
- C. Use identical TTL expiry for all hot keys
- D. Cache personalized responses under one global key

Q280**EXPERT**

In cloud architecture, signed URLs are required. Which conclusion or design is best?

- A. A CDN removes the need to secure private content
- B. Use identical TTL expiry for all hot keys
- C. they provide time-limited access when validation is enforced at the edge or origin
- D. Cache personalized responses under one global key

Q281**INTERMEDIATE**

In cloud architecture, an IAM policy grants * on *. Which conclusion or design is best?

- A. Grant wildcard permissions and rely on audit logs
- B. replace it with required actions and resources
- C. Use permanent administrator keys for automation
- D. Trust policy conditions are unnecessary

Q282**ADVANCED**

In cloud architecture, a role is assumed by workloads. Which conclusion or design is best?

- A. use conditions to restrict trusted principals and context
- B. Use permanent administrator keys for automation
- C. Trust policy conditions are unnecessary
- D. Grant wildcard permissions and rely on audit logs

Q283**ADVANCED**

In cloud architecture, permissions are unused for 90 days. Which conclusion or design is best?

- A. review and remove them after validating operational needs
- B. Trust policy conditions are unnecessary
- C. Grant wildcard permissions and rely on audit logs
- D. Use permanent administrator keys for automation

Q284**EXPERT**

In cloud architecture, an administrator uses a permanent access key. Which conclusion or design is best?

- A. Use permanent administrator keys for automation
- B. prefer federated human access with MFA and short sessions
- C. Grant wildcard permissions and rely on audit logs
- D. Trust policy conditions are unnecessary

Q285**INTERMEDIATE**

In cloud architecture, RPO is 15 minutes. Which conclusion or design is best?

- A. the design may lose at most about 15 minutes of data after a disaster
- B. Nightly backups meet any RPO below 24 hours
- C. RTO measures acceptable data loss
- D. RPO measures recovery duration

Q286**ADVANCED**

In cloud architecture, RTO is two hours. Which conclusion or design is best?

- A. RPO measures recovery duration
- B. RTO measures acceptable data loss
- C. Nightly backups meet any RPO below 24 hours
- D. service should be restored within two hours of the incident target

Q287**ADVANCED**

In cloud architecture, backups run nightly but RPO is one hour. Which conclusion or design is best?

- A. RTO measures acceptable data loss
- B. the backup design does not meet the objective
- C. RPO measures recovery duration
- D. Nightly backups meet any RPO below 24 hours

Q288**EXPERT**

In cloud architecture, a failover plan meets RTO only when staff are available. Which conclusion or design is best?

- A. automate and test the recovery path
- B. RTO measures acceptable data loss
- C. RPO measures recovery duration
- D. Nightly backups meet any RPO below 24 hours

Q289**INTERMEDIATE**

In cloud architecture, steady baseline compute runs all year. Which conclusion or design is best?

- A. commitment discounts may reduce cost if sizing is understood
- B. Use interruptible instances for a critical singleton
- C. Reserved pricing is always cheaper for bursty uncertain workloads
- D. Buy long commitments before rightsizing

Q290

ADVANCED

In cloud architecture, fault-tolerant batch jobs can be interrupted. Which conclusion or design is best?

- A. Use interruptible instances for a critical singleton
- B. spot or preemptible capacity is suitable
- C. Buy long commitments before rightsizing
- D. Reserved pricing is always cheaper for bursty uncertain workloads

Q291

ADVANCED

In cloud architecture, a critical singleton database uses spot instances. Which conclusion or design is best?

- A. Reserved pricing is always cheaper for bursty uncertain workloads
- B. Use interruptible instances for a critical singleton
- C. this creates unacceptable interruption risk
- D. Buy long commitments before rightsizing

Q292

EXPERT

In cloud architecture, commitments are bought before rightsizing. Which conclusion or design is best?

- A. Use interruptible instances for a critical singleton
- B. Reserved pricing is always cheaper for bursty uncertain workloads
- C. the organization may lock in waste
- D. Buy long commitments before rightsizing

Q293

INTERMEDIATE

In cloud architecture, data crosses regions frequently. Which conclusion or design is best?

- A. network egress charges may be significant
- B. Observability export volume does not affect egress
- C. NAT data processing has no cost
- D. Inter-region transfer is always free

Q294

ADVANCED

In cloud architecture, an application repeatedly downloads the same object from another region. Which conclusion or design is best?

- A. Inter-region transfer is always free
- B. Observability export volume does not affect egress
- C. replication or caching may lower latency and transfer cost
- D. NAT data processing has no cost

Q295**ADVANCED**

In cloud architecture, NAT gateway data processing is high. Which conclusion or design is best?

- A. Observability export volume does not affect egress
- B. Inter-region transfer is always free
- C. review traffic paths and private endpoints
- D. NAT data processing has no cost

Q296**EXPERT**

In cloud architecture, logs are exported to an external provider. Which conclusion or design is best?

- A. Observability export volume does not affect egress
- B. include sustained egress in observability cost planning
- C. Inter-region transfer is always free
- D. NAT data processing has no cost

Q297**INTERMEDIATE**

In cloud architecture, producers should not block when consumers are temporarily slow. Which conclusion or design is best?

- A. Retry poison messages forever
- B. insert a durable queue
- C. Require global ordering for every message
- D. Assume exactly-once processing without idempotency

Q298**ADVANCED**

In cloud architecture, messages may be delivered more than once. Which conclusion or design is best?

- A. Assume exactly-once processing without idempotency
- B. Require global ordering for every message
- C. Retry poison messages forever
- D. make consumers idempotent

Q299**ADVANCED**

In cloud architecture, message order is required per customer. Which conclusion or design is best?

- A. use partitioning or FIFO semantics keyed by customer
- B. Assume exactly-once processing without idempotency
- C. Require global ordering for every message
- D. Retry poison messages forever

Q300**EXPERT**

In cloud architecture, a poison message retries forever. Which conclusion or design is best?

- A. Require global ordering for every message
- B. Assume exactly-once processing without idempotency
- C. Retry poison messages forever
- D. use retry limits and a dead-letter queue

Q301**INTERMEDIATE**

In cloud architecture, private instances need outbound internet access. Which conclusion or design is best?

- A. Overlapping CIDRs peer without issue
- B. route through a NAT service without public inbound exposure
- C. An internet-gateway route alone gives private addresses public reachability
- D. Use public NAT even when a private service endpoint exists

Q302**ADVANCED**

In cloud architecture, private workloads access cloud object storage through public NAT. Which conclusion or design is best?

- A. Overlapping CIDRs peer without issue
- B. An internet-gateway route alone gives private addresses public reachability
- C. a private service endpoint can improve security and reduce NAT cost
- D. Use public NAT even when a private service endpoint exists

Q303**ADVANCED**

In cloud architecture, two private networks have overlapping CIDRs. Which conclusion or design is best?

- A. An internet-gateway route alone gives private addresses public reachability
- B. Use public NAT even when a private service endpoint exists
- C. simple peering cannot route them cleanly
- D. Overlapping CIDRs peer without issue

Q304**EXPERT**

In cloud architecture, a route table sends 0.0.0.0/0 to an internet gateway. Which conclusion or design is best?

- A. An internet-gateway route alone gives private addresses public reachability
- B. only resources with suitable public addressing and policy can use it directly
- C. Use public NAT even when a private service endpoint exists
- D. Overlapping CIDRs peer without issue

Q305

INTERMEDIATE

In cloud architecture, a serverless function has a 15-minute maximum runtime. Which conclusion or design is best?

- A. Assume retried events cannot duplicate side effects
- B. Ignore runtime and concurrency limits
- C. longer jobs need decomposition or another compute service
- D. Open one database connection per invocation without pooling

Q306

ADVANCED

In cloud architecture, cold starts affect latency. Which conclusion or design is best?

- A. Assume retried events cannot duplicate side effects
- B. provisioned concurrency, smaller packages, or asynchronous patterns can help
- C. Ignore runtime and concurrency limits
- D. Open one database connection per invocation without pooling

Q307

ADVANCED

In cloud architecture, thousands of functions open database connections. Which conclusion or design is best?

- A. Open one database connection per invocation without pooling
- B. use pooling or a managed proxy
- C. Assume retried events cannot duplicate side effects
- D. Ignore runtime and concurrency limits

Q308

EXPERT

In cloud architecture, an event is retried after timeout. Which conclusion or design is best?

- A. Open one database connection per invocation without pooling
- B. the function must be idempotent
- C. Assume retried events cannot duplicate side effects
- D. Ignore runtime and concurrency limits

Q309

INTERMEDIATE

In cloud architecture, data is encrypted with a managed KMS key. Which conclusion or design is best?

- A. Disable a key without impact to applications
- B. Use one unrestricted key for all environments
- C. access to both ciphertext and key permissions is required
- D. Anyone with ciphertext can decrypt without key permission

Q310

ADVANCED

In cloud architecture, a key is disabled. Which conclusion or design is best?

- A. dependent decrypt operations fail
- B. Disable a key without impact to applications
- C. Use one unrestricted key for all environments
- D. Anyone with ciphertext can decrypt without key permission

Q311

ADVANCED

In cloud architecture, key rotation is enabled. Which conclusion or design is best?

- A. Use one unrestricted key for all environments
- B. new cryptographic material is used while old material remains for decryption
- C. Disable a key without impact to applications
- D. Anyone with ciphertext can decrypt without key permission

Q312

EXPERT

In cloud architecture, one key protects every environment. Which conclusion or design is best?

- A. separate keys and policies reduce blast radius
- B. Disable a key without impact to applications
- C. Use one unrestricted key for all environments
- D. Anyone with ciphertext can decrypt without key permission

Monitoring and Observability

MONITORING AND OBSERVABILITY

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Monitoring and Observability Questions

Q313**INTERMEDIATE**

For monitoring and observability, an alert uses absolute 5xx requests per second on a high-traffic service. What is the best answer?

- A. Use only the absolute error count
- B. Average every route into one value regardless of traffic
- C. alert on the 5xx ratio to total requests when percentage is the objective
- D. Ignore the total request denominator

Q314**ADVANCED**

For monitoring and observability, 5xx rate is 20 rps out of 10,000 rps. What is the best answer?

- A. the error ratio is 0.2%
- B. Ignore the total request denominator
- C. Average every route into one value regardless of traffic
- D. Use only the absolute error count

Q315**ADVANCED**

For monitoring and observability, a low-traffic service has one failure out of one request. What is the best answer?

- A. Average every route into one value regardless of traffic
- B. Use only the absolute error count
- C. the ratio is high but needs minimum-volume handling
- D. Ignore the total request denominator

Q316**EXPERT**

For monitoring and observability, errors are grouped by route. What is the best answer?

- A. aggregate carefully so one failing route is not hidden by global traffic
- B. Ignore the total request denominator
- C. Use only the absolute error count
- D. Average every route into one value regardless of traffic

Q317

INTERMEDIATE

For monitoring and observability, a Prometheus counter is sampled over five minutes. What is the best answer?

- A. Treat a counter as a gauge
- B. rate estimates per-second growth
- C. Use irate for every long-window alert
- D. Subtract raw counters without handling resets

Q318

ADVANCED

For monitoring and observability, you want the approximate number of requests in one hour. What is the best answer?

- A. Use irate for every long-window alert
- B. Subtract raw counters without handling resets
- C. use increase over one hour
- D. Treat a counter as a gauge

Q319

ADVANCED

For monitoring and observability, irate is used for a slow-moving alert. What is the best answer?

- A. rate is usually more stable
- B. Subtract raw counters without handling resets
- C. Treat a counter as a gauge
- D. Use irate for every long-window alert

Q320

EXPERT

For monitoring and observability, a counter resets after restart. What is the best answer?

- A. Subtract raw counters without handling resets
- B. Treat a counter as a gauge
- C. rate and increase account for the reset
- D. Use irate for every long-window alert

Q321

INTERMEDIATE

For monitoring and observability, a metric label contains user_id. What is the best answer?

- A. cardinality can explode
- B. Use raw UUID paths as metric labels
- C. Increase retention to solve cardinality
- D. Label metrics with every user and request ID

Q322

ADVANCED

For monitoring and observability, every request path includes a unique UUID label. What is the best answer?

- A. Increase retention to solve cardinality
- B. Label metrics with every user and request ID
- C. normalize the route label
- D. Use raw UUID paths as metric labels

Q323

ADVANCED

For monitoring and observability, Prometheus memory grows with active series. What is the best answer?

- A. reduce label cardinality and retention pressure
- B. Use raw UUID paths as metric labels
- C. Label metrics with every user and request ID
- D. Increase retention to solve cardinality

Q324

EXPERT

For monitoring and observability, debug labels are useful briefly. What is the best answer?

- A. Increase retention to solve cardinality
- B. send them to logs or controlled exemplars rather than permanent metrics
- C. Label metrics with every user and request ID
- D. Use raw UUID paths as metric labels

Q325

INTERMEDIATE

For monitoring and observability, request duration is a histogram. What is the best answer?

- A. Store every request as a metric series
- B. Use the +Inf bucket alone for accurate p95
- C. Sum client-side quantiles across instances
- D. histogram_quantile can estimate p95 from bucket rates

Q326

ADVANCED

For monitoring and observability, a summary reports client-side quantiles. What is the best answer?

- A. quantiles generally cannot be aggregated correctly across instances
- B. Store every request as a metric series
- C. Sum client-side quantiles across instances
- D. Use the +Inf bucket alone for accurate p95

Q327

ADVANCED

For monitoring and observability, all histogram observations fall in the +Inf bucket. What is the best answer?

- A. Use the +Inf bucket alone for accurate p95
- B. Sum client-side quantiles across instances
- C. Store every request as a metric series
- D. bucket boundaries are poorly chosen for useful quantiles

Q328

EXPERT

For monitoring and observability, you need exact per-request latency. What is the best answer?

- A. Store every request as a metric series
- B. use traces or logs, not aggregated histograms
- C. Use the +Inf bucket alone for accurate p95
- D. Sum client-side quantiles across instances

Q329

INTERMEDIATE

For monitoring and observability, an alert fires daily but no action is taken. What is the best answer?

- A. Route all severities to the same pager
- B. Page on alerts that require no action
- C. Keep duplicate symptom alerts uninhibited
- D. remove, tune, or assign an actionable response

Q330

ADVANCED

For monitoring and observability, warning and critical alerts page the same channel. What is the best answer?

- A. Page on alerts that require no action
- B. Keep duplicate symptom alerts uninhibited
- C. Route all severities to the same pager
- D. route by urgency and required response

Q331

ADVANCED

For monitoring and observability, one incident triggers 200 dependent alerts. What is the best answer?

- A. Keep duplicate symptom alerts uninhibited
- B. Page on alerts that require no action
- C. use inhibition, grouping, and root-cause alerting
- D. Route all severities to the same pager

Q332

EXPERT

For monitoring and observability, an alert has no runbook or owner. What is the best answer?

- A. Route all severities to the same pager
- B. add ownership, context, and a response procedure
- C. Keep duplicate symptom alerts uninhibited
- D. Page on alerts that require no action

Q333

INTERMEDIATE

For monitoring and observability, availability is defined as successful eligible requests divided by total eligible requests. What is the best answer?

- A. this is an SLI
- B. Define availability without eligible-event rules
- C. Call an SLO a legal penalty contract
- D. Treat an internal process metric as the user SLI

Q334

ADVANCED

For monitoring and observability, the target is 99.9% monthly availability. What is the best answer?

- A. Treat an internal process metric as the user SLI
- B. Call an SLO a legal penalty contract
- C. this is an SLO
- D. Define availability without eligible-event rules

Q335

ADVANCED

For monitoring and observability, a provider contract includes penalties below 99.9%. What is the best answer?

- A. this is an SLA
- B. Call an SLO a legal penalty contract
- C. Define availability without eligible-event rules
- D. Treat an internal process metric as the user SLI

Q336

EXPERT

For monitoring and observability, internal health checks are perfect but users fail. What is the best answer?

- A. Treat an internal process metric as the user SLI
- B. Call an SLO a legal penalty contract
- C. the SLI is not measuring the user experience
- D. Define availability without eligible-event rules

Q337

INTERMEDIATE

For monitoring and observability, a service consumes its monthly error budget in two hours. What is the best answer?

- A. Ignore persistent slow budget consumption
- B. Use error budget only as a dashboard decoration
- C. a fast burn-rate alert should page
- D. Alert only when the monthly window is already over

Q338

ADVANCED

For monitoring and observability, a small persistent issue consumes budget over days. What is the best answer?

- A. a slower burn window should create a lower-urgency alert
- B. Ignore persistent slow budget consumption
- C. Alert only when the monthly window is already over
- D. Use error budget only as a dashboard decoration

Q339

ADVANCED

For monitoring and observability, an SLO is met with lots of unused budget. What is the best answer?

- A. Ignore persistent slow budget consumption
- B. Use error budget only as a dashboard decoration
- C. Alert only when the monthly window is already over
- D. the team may accept more release risk within policy

Q340

EXPERT

For monitoring and observability, the budget is exhausted. What is the best answer?

- A. Use error budget only as a dashboard decoration
- B. Ignore persistent slow budget consumption
- C. freeze risky changes and prioritize reliability according to policy
- D. Alert only when the monthly window is already over

Q341

INTERMEDIATE

For monitoring and observability, logs are plain text with inconsistent fields. What is the best answer?

- A. emit structured logs with stable keys
- B. Emit passwords and redact them later
- C. Log unstructured payloads with inconsistent fields
- D. Record every large successful payload indefinitely

Q342

ADVANCED

For monitoring and observability, a request ID appears in every service log. What is the best answer?

- A. Log unstructured payloads with inconsistent fields
- B. cross-service investigation becomes easier
- C. Record every large successful payload indefinitely
- D. Emit passwords and redact them later

Q343

ADVANCED

For monitoring and observability, passwords are logged for debugging. What is the best answer?

- A. remove and rotate exposed credentials
- B. Emit passwords and redact them later
- C. Record every large successful payload indefinitely
- D. Log unstructured payloads with inconsistent fields

Q344

EXPERT

For monitoring and observability, every successful request logs a large payload. What is the best answer?

- A. Log unstructured payloads with inconsistent fields
- B. Record every large successful payload indefinitely
- C. Emit passwords and redact them later
- D. sampling and field selection may be needed

Q345

INTERMEDIATE

For monitoring and observability, a trace lacks propagation across an async queue. What is the best answer?

- A. Use random sampling with no protection for rare errors
- B. Drop trace context at queue boundaries
- C. carry trace context in message metadata
- D. Keep logs and traces unrelated

Q346

ADVANCED

For monitoring and observability, sampling is 1%. What is the best answer?

- A. Keep logs and traces unrelated
- B. Use random sampling with no protection for rare errors
- C. Drop trace context at queue boundaries
- D. rare failures may be missed unless tail-based or error-biased sampling is used

Q347

ADVANCED

For monitoring and observability, one span dominates trace latency. What is the best answer?

- A. Use random sampling with no protection for rare errors
- B. Keep logs and traces unrelated
- C. Drop trace context at queue boundaries
- D. investigate that operation and its downstream dependency

Q348

EXPERT

For monitoring and observability, trace IDs are included in logs. What is the best answer?

- A. engineers can pivot between traces and detailed events
- B. Drop trace context at queue boundaries
- C. Use random sampling with no protection for rare errors
- D. Keep logs and traces unrelated

Q349

INTERMEDIATE

For monitoring and observability, Prometheus target is down. What is the best answer?

- A. Assume up=0 proves the application is unavailable
- B. Make scrape duration longer than the interval
- C. Restart the application before checking scrape configuration
- D. check service discovery, network reachability, TLS/auth, and exporter health

Q350

ADVANCED

For monitoring and observability, up is 0 but the application serves traffic. What is the best answer?

- A. Make scrape duration longer than the interval
- B. Assume up=0 proves the application is unavailable
- C. Restart the application before checking scrape configuration
- D. the metrics endpoint or scrape configuration may be broken

Q351

ADVANCED

For monitoring and observability, scrape duration approaches the scrape interval. What is the best answer?

- A. Assume up=0 proves the application is unavailable
- B. collection may overlap or time out
- C. Make scrape duration longer than the interval
- D. Restart the application before checking scrape configuration

Q352

EXPERT

For monitoring and observability, a target disappears from discovery. What is the best answer?

- A. Assume up=0 proves the application is unavailable
- B. Restart the application before checking scrape configuration
- C. Make scrape duration longer than the interval
- D. inspect labels and discovery configuration before the exporter

Q353

INTERMEDIATE

For monitoring and observability, a raw counter graph drops to zero after restart. What is the best answer?

- A. Treat a reset as negative traffic
- B. Use a counter for naturally decreasing state
- C. this is a counter reset
- D. Graph raw counters across restarts as continuous values

Q354

ADVANCED

For monitoring and observability, you sum rates across instances. What is the best answer?

- A. Graph raw counters across restarts as continuous values
- B. this is valid for total throughput
- C. Treat a reset as negative traffic
- D. Use a counter for naturally decreasing state

Q355

ADVANCED

For monitoring and observability, you sum raw counters with different restart times. What is the best answer?

- A. Graph raw counters across restarts as continuous values
- B. Treat a reset as negative traffic
- C. the graph can be misleading
- D. Use a counter for naturally decreasing state

Q356

EXPERT

For monitoring and observability, a metric can decrease naturally. What is the best answer?

- A. Graph raw counters across restarts as continuous values
- B. Treat a reset as negative traffic
- C. use a gauge rather than a counter
- D. Use a counter for naturally decreasing state

Q357

INTERMEDIATE

For monitoring and observability, a dashboard repeats an expensive query many times. What is the best answer?

- A. Repeat expensive queries in every dashboard panel
- B. create a recording rule
- C. Use an evaluation interval too coarse for the signal
- D. Create recorded series with unbounded labels

Q358

ADVANCED

For monitoring and observability, an alert expression is complex and reused. What is the best answer?

- A. a named recording rule improves consistency
- B. Use an evaluation interval too coarse for the signal
- C. Repeat expensive queries in every dashboard panel
- D. Create recorded series with unbounded labels

Q359

ADVANCED

For monitoring and observability, a recording rule interval is too coarse. What is the best answer?

- A. short spikes may be smoothed or missed
- B. Repeat expensive queries in every dashboard panel
- C. Create recorded series with unbounded labels
- D. Use an evaluation interval too coarse for the signal

Q360

EXPERT

For monitoring and observability, recorded series add labels carelessly. What is the best answer?

- A. Repeat expensive queries in every dashboard panel
- B. Use an evaluation interval too coarse for the signal
- C. they can increase cardinality
- D. Create recorded series with unbounded labels

Q361

INTERMEDIATE

For monitoring and observability, you probe the public HTTPS endpoint from outside the cluster. What is the best answer?

- A. Rely only on black-box checks for root cause
- B. this is black-box monitoring
- C. Assume internal metrics prove the public path works
- D. Probe only from inside the same process

Q362**ADVANCED**

For monitoring and observability, you inspect queue depth and thread pools. What is the best answer?

- A. this is white-box monitoring
- B. Probe only from inside the same process
- C. Rely only on black-box checks for root cause
- D. Assume internal metrics prove the public path works

Q363**ADVANCED**

For monitoring and observability, internal metrics are healthy but the public probe fails. What is the best answer?

- A. Rely only on black-box checks for root cause
- B. Probe only from inside the same process
- C. Assume internal metrics prove the public path works
- D. investigate DNS, CDN, load balancer, TLS, and network path

Q364**EXPERT**

For monitoring and observability, only black-box checks exist. What is the best answer?

- A. Assume internal metrics prove the public path works
- B. Probe only from inside the same process
- C. Rely only on black-box checks for root cause
- D. add internal metrics for diagnosis

Networking

NETWORKING

Questions 365-416 | 52 questions

Networking Questions

Q365**INTERMEDIATE**

In network troubleshooting, a TCP client sends SYN and receives SYN-ACK. What is the correct interpretation or action?

- A. A SYN alone means the connection is established
- B. A timeout is identical to an immediate RST
- C. Application success is proven by the TCP handshake
- D. it should send ACK to complete the handshake

Q366**ADVANCED**

In network troubleshooting, SYN packets are retransmitted with no reply. What is the correct interpretation or action?

- A. A SYN alone means the connection is established
- B. Application success is proven by the TCP handshake
- C. the path, firewall, or listener may be dropping them
- D. A timeout is identical to an immediate RST

Q367**ADVANCED**

In network troubleshooting, RST is returned immediately. What is the correct interpretation or action?

- A. A SYN alone means the connection is established
- B. the destination actively rejected the connection or no service is listening
- C. Application success is proven by the TCP handshake
- D. A timeout is identical to an immediate RST

Q368**EXPERT**

In network troubleshooting, the server accepts a connection then closes it after application data. What is the correct interpretation or action?

- A. A timeout is identical to an immediate RST
- B. Application success is proven by the TCP handshake
- C. A SYN alone means the connection is established
- D. the failure is above the initial TCP handshake

Q369

INTERMEDIATE

In network troubleshooting, a DNS resolver returns NXDOMAIN. What is the correct interpretation or action?

- A. TTL changes invalidate every client cache instantly
- B. NXDOMAIN means the DNS server timed out
- C. the queried name does not exist according to the authoritative chain
- D. SERVFAIL proves the name does not exist

Q370

ADVANCED

In network troubleshooting, SERVFAIL is returned. What is the correct interpretation or action?

- A. NXDOMAIN means the DNS server timed out
- B. TTL changes invalidate every client cache instantly
- C. SERVFAIL proves the name does not exist
- D. the resolver could not complete resolution successfully

Q371

ADVANCED

In network troubleshooting, a record changed but clients still use the old address. What is the correct interpretation or action?

- A. TTL changes invalidate every client cache instantly
- B. cached TTL may not have expired
- C. NXDOMAIN means the DNS server timed out
- D. SERVFAIL proves the name does not exist

Q372

EXPERT

In network troubleshooting, an internal and public zone use the same domain. What is the correct interpretation or action?

- A. split-horizon DNS may return different answers by resolver location
- B. NXDOMAIN means the DNS server timed out
- C. TTL changes invalidate every client cache instantly
- D. SERVFAIL proves the name does not exist

Q373

INTERMEDIATE

In network troubleshooting, a reverse proxy returns 502. What is the correct interpretation or action?

- A. 504 means no route matched
- B. 503 always proves a DNS failure
- C. 502 always means the client closed the request
- D. it received an invalid response or connection failure from the upstream

Q374

ADVANCED

In network troubleshooting, a reverse proxy returns 504. What is the correct interpretation or action?

- A. 502 always means the client closed the request
- B. the upstream did not respond within the configured time
- C. 504 means no route matched
- D. 503 always proves a DNS failure

Q375

ADVANCED

In network troubleshooting, the application returns 503 intentionally during overload. What is the correct interpretation or action?

- A. the service is unavailable and may include Retry-After
- B. 503 always proves a DNS failure
- C. 504 means no route matched
- D. 502 always means the client closed the request

Q376

EXPERT

In network troubleshooting, clients receive 499 in NGINX logs. What is the correct interpretation or action?

- A. 503 always proves a DNS failure
- B. 504 means no route matched
- C. the client closed the request before the server completed it
- D. 502 always means the client closed the request

Q377

INTERMEDIATE

In network troubleshooting, a certificate is valid but issued for api.example.com while the client requests db.example.com. What is the correct interpretation or action?

- A. SNI is unrelated to certificate selection
- B. hostname verification fails
- C. The server never needs to send intermediate certificates
- D. Accept any certificate with a valid date regardless of hostname

Q378

ADVANCED

In network troubleshooting, the certificate chain omits an intermediate. What is the correct interpretation or action?

- A. Accept any certificate with a valid date regardless of hostname
- B. The server never needs to send intermediate certificates
- C. some clients cannot build trust
- D. SNI is unrelated to certificate selection

Q379**ADVANCED**

In network troubleshooting, TLS fails only on old clients. What is the correct interpretation or action?

- A. Accept any certificate with a valid date regardless of hostname
- B. SNI is unrelated to certificate selection
- C. The server never needs to send intermediate certificates
- D. protocol or cipher compatibility may be the cause

Q380**EXPERT**

In network troubleshooting, SNI is missing from a client connecting to a shared IP. What is the correct interpretation or action?

- A. SNI is unrelated to certificate selection
- B. The server never needs to send intermediate certificates
- C. the server may present the wrong certificate
- D. Accept any certificate with a valid date regardless of hostname

Q381**INTERMEDIATE**

In network troubleshooting, 10.0.0.0/24 is used. What is the correct interpretation or action?

- A. A /24 has 24 total addresses
- B. A /32 represents an entire subnet of 32 hosts
- C. it contains 256 addresses from .0 through .255
- D. 10.0.0.0/24 and 10.0.0.128/25 are disjoint

Q382**ADVANCED**

In network troubleshooting, two subnets are 10.0.0.0/24 and 10.0.0.128/25. What is the correct interpretation or action?

- A. 10.0.0.0/24 and 10.0.0.128/25 are disjoint
- B. they overlap
- C. A /24 has 24 total addresses
- D. A /32 represents an entire subnet of 32 hosts

Q383**ADVANCED**

In network troubleshooting, a /30 IPv4 network is used traditionally. What is the correct interpretation or action?

- A. A /32 represents an entire subnet of 32 hosts
- B. A /24 has 24 total addresses
- C. 10.0.0.0/24 and 10.0.0.128/25 are disjoint
- D. it provides two usable host addresses

Q384**EXPERT**

In network troubleshooting, a route has /32. What is the correct interpretation or action?

- A. 10.0.0.0/24 and 10.0.0.128/25 are disjoint
- B. it matches one IPv4 address
- C. A /32 represents an entire subnet of 32 hosts
- D. A /24 has 24 total addresses

Q385**INTERMEDIATE**

In network troubleshooting, small packets work but large packets hang through a tunnel. What is the correct interpretation or action?

- A. Block all ICMP to improve PMTUD
- B. suspect MTU or path-MTU discovery issues
- C. Increase application memory to solve MTU failure
- D. Enable jumbo frames on only one interface

Q386**ADVANCED**

In network troubleshooting, ICMP fragmentation-needed messages are blocked. What is the correct interpretation or action?

- A. Block all ICMP to improve PMTUD
- B. Increase application memory to solve MTU failure
- C. PMTUD can fail
- D. Enable jumbo frames on only one interface

Q387**ADVANCED**

In network troubleshooting, TCP MSS clamping is configured at a tunnel edge. What is the correct interpretation or action?

- A. Increase application memory to solve MTU failure
- B. Enable jumbo frames on only one interface
- C. it can avoid oversized TCP segments
- D. Block all ICMP to improve PMTUD

Q388**EXPERT**

In network troubleshooting, jumbo frames are enabled on only part of a path. What is the correct interpretation or action?

- A. Block all ICMP to improve PMTUD
- B. Enable jumbo frames on only one interface
- C. Increase application memory to solve MTU failure
- D. inconsistent MTU can cause drops or fragmentation

Q389**INTERMEDIATE**

In network troubleshooting, many private hosts share one public IPv4 address. What is the correct interpretation or action?

- A. Outbound source NAT automatically publishes inbound services
- B. PAT maps flows using ports
- C. Embedded private addresses are always translated correctly
- D. NAT never keeps connection state

Q390**ADVANCED**

In network troubleshooting, an inbound connection targets a private service through NAT. What is the correct interpretation or action?

- A. Outbound source NAT automatically publishes inbound services
- B. a destination translation or port-forward rule is required
- C. NAT never keeps connection state
- D. Embedded private addresses are always translated correctly

Q391**ADVANCED**

In network troubleshooting, NAT state expires during a long idle session. What is the correct interpretation or action?

- A. Embedded private addresses are always translated correctly
- B. NAT never keeps connection state
- C. Outbound source NAT automatically publishes inbound services
- D. keepalives or timeout tuning may be needed

Q392

EXPERT

In network troubleshooting, an application embeds private IPs in payloads. What is the correct interpretation or action?

- A. NAT may not transparently fix the protocol
- B. Outbound source NAT automatically publishes inbound services
- C. NAT never keeps connection state
- D. Embedded private addresses are always translated correctly

Q393

INTERMEDIATE

In network troubleshooting, routing must use HTTP paths and headers. What is the correct interpretation or action?

- A. Use L4 balancing for HTTP path routing
- B. TLS termination never changes trust boundaries
- C. use a layer-7 load balancer
- D. Source IP is always preserved through NAT

Q394

ADVANCED

In network troubleshooting, millions of raw TCP connections need minimal overhead. What is the correct interpretation or action?

- A. TLS termination never changes trust boundaries
- B. a layer-4 load balancer is often appropriate
- C. Use L4 balancing for HTTP path routing
- D. Source IP is always preserved through NAT

Q395

ADVANCED

In network troubleshooting, TLS terminates at the load balancer. What is the correct interpretation or action?

- A. Source IP is always preserved through NAT
- B. backend traffic may be plaintext or re-encrypted according to policy
- C. TLS termination never changes trust boundaries
- D. Use L4 balancing for HTTP path routing

Q396

EXPERT

In network troubleshooting, source IP must be preserved. What is the correct interpretation or action?

- A. Source IP is always preserved through NAT
- B. TLS termination never changes trust boundaries
- C. use a supported proxy protocol, headers, or direct-routing design
- D. Use L4 balancing for HTTP path routing

Q397

INTERMEDIATE

In network troubleshooting, many sockets remain in `TIME_WAIT` on the active closer. What is the correct interpretation or action?

- A. `SYN_RECV` means every handshake completed
- B. `CLOSE_WAIT` is fixed by lowering the TCP timeout only
- C. this is normal TCP protection against delayed segments
- D. `TIME_WAIT` always indicates an application leak

Q398

ADVANCED

In network troubleshooting, many sockets remain in `CLOSE_WAIT`. What is the correct interpretation or action?

- A. `SYN_RECV` means every handshake completed
- B. the local application has not closed after the peer closed
- C. `CLOSE_WAIT` is fixed by lowering the TCP timeout only
- D. `TIME_WAIT` always indicates an application leak

Q399

ADVANCED

In network troubleshooting, `SYN_RECV` grows rapidly. What is the correct interpretation or action?

- A. the server has many half-open handshakes
- B. `SYN_RECV` means every handshake completed
- C. `TIME_WAIT` always indicates an application leak
- D. `CLOSE_WAIT` is fixed by lowering the TCP timeout only

Q400

EXPERT

In network troubleshooting, `ESTABLISHED` exists but no traffic moves. What is the correct interpretation or action?

- A. inspect application blocking, window sizes, and packet flow
- B. `TIME_WAIT` always indicates an application leak
- C. `SYN_RECV` means every handshake completed
- D. `CLOSE_WAIT` is fixed by lowering the TCP timeout only

Q401**INTERMEDIATE**

In network troubleshooting, two routes match and one has a longer prefix. What is the correct interpretation or action?

- A. A default route is unnecessary for unknown destinations
- B. the longest-prefix route wins
- C. The least-specific route always wins
- D. Stateful firewalls are unaffected by asymmetric routing

Q402**ADVANCED**

In network troubleshooting, prefix lengths are equal but metrics differ. What is the correct interpretation or action?

- A. A default route is unnecessary for unknown destinations
- B. the lower metric is generally preferred
- C. Stateful firewalls are unaffected by asymmetric routing
- D. The least-specific route always wins

Q403**ADVANCED**

In network troubleshooting, the default route is absent. What is the correct interpretation or action?

- A. Stateful firewalls are unaffected by asymmetric routing
- B. non-local destinations without specific routes are unreachable
- C. A default route is unnecessary for unknown destinations
- D. The least-specific route always wins

Q404**EXPERT**

In network troubleshooting, asymmetric routing crosses a stateful firewall. What is the correct interpretation or action?

- A. A default route is unnecessary for unknown destinations
- B. Stateful firewalls are unaffected by asymmetric routing
- C. return traffic may be dropped
- D. The least-specific route always wins

Q405

INTERMEDIATE

In network troubleshooting, a broad ACCEPT rule appears before a later DROP rule. What is the correct interpretation or action?

- A. A zero counter proves the rule is correct
- B. the later rule may never match
- C. Apply remote firewall changes without rollback access
- D. Firewall rule order never matters

Q406

ADVANCED

In network troubleshooting, established connections should be permitted. What is the correct interpretation or action?

- A. Firewall rule order never matters
- B. Apply remote firewall changes without rollback access
- C. A zero counter proves the rule is correct
- D. match ESTABLISHED,RELATED state before restrictive new-connection rules

Q407

ADVANCED

In network troubleshooting, a policy change locks out SSH. What is the correct interpretation or action?

- A. Firewall rule order never matters
- B. use an out-of-band path or atomic rollback procedure
- C. Apply remote firewall changes without rollback access
- D. A zero counter proves the rule is correct

Q408

EXPERT

In network troubleshooting, packet counters on a rule remain zero. What is the correct interpretation or action?

- A. A zero counter proves the rule is correct
- B. Apply remote firewall changes without rollback access
- C. traffic may not traverse that chain or match its conditions
- D. Firewall rule order never matters

Q409

INTERMEDIATE

In network troubleshooting, a load balancer closes idle connections after 60 seconds. What is the correct interpretation or action?

- A. Set every timeout to infinite
- B. application keepalives or timeout alignment may be needed
- C. Disable HTTP connection reuse
- D. Use aggressive keepalives without scale analysis

Q410

ADVANCED

In network troubleshooting, HTTP keep-alive is enabled. What is the correct interpretation or action?

- A. Disable HTTP connection reuse
- B. Set every timeout to infinite
- C. Use aggressive keepalives without scale analysis
- D. multiple requests can reuse one TCP connection

Q411

ADVANCED

In network troubleshooting, TCP keepalive detects dead peers slowly by default. What is the correct interpretation or action?

- A. Use aggressive keepalives without scale analysis
- B. Disable HTTP connection reuse
- C. tune it only with awareness of network and scale impact
- D. Set every timeout to infinite

Q412

EXPERT

In network troubleshooting, a proxy timeout is shorter than the backend's legitimate response time. What is the correct interpretation or action?

- A. Disable HTTP connection reuse
- B. clients receive premature gateway errors
- C. Set every timeout to infinite
- D. Use aggressive keepalives without scale analysis

Q413

INTERMEDIATE

In network troubleshooting, you need to confirm packets reach interface eth0 on port 443. What is the correct interpretation or action?

- A. Capture location does not affect interpretation
- B. No SYN-ACK proves the client never sent SYN
- C. capture with tcpdump using a focused filter
- D. A packet capture automatically decrypts TLS payloads

Q414

ADVANCED

In network troubleshooting, tcpdump sees SYN but not SYN-ACK. What is the correct interpretation or action?

- A. A packet capture automatically decrypts TLS payloads
- B. inspect the local listener and firewall
- C. Capture location does not affect interpretation
- D. No SYN-ACK proves the client never sent SYN

Q415

ADVANCED

In network troubleshooting, tcpdump on any sees duplicate-looking packets. What is the correct interpretation or action?

- A. Capture location does not affect interpretation
- B. Linux may show packets on multiple logical interfaces
- C. A packet capture automatically decrypts TLS payloads
- D. No SYN-ACK proves the client never sent SYN

Q416

EXPERT

In network troubleshooting, payload is encrypted with TLS. What is the correct interpretation or action?

- A. packet capture still shows timing, endpoints, and handshake metadata but not plaintext
- B. Capture location does not affect interpretation
- C. No SYN-ACK proves the client never sent SYN
- D. A packet capture automatically decrypts TLS payloads

Security

SECURITY

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Security Questions

Q417**INTERMEDIATE**

From a security perspective, a service account can administer the whole cluster but only reads one ConfigMap. What is the strongest response?

- A. Retain stale access indefinitely
- B. Keep broad permissions and review logs later
- C. reduce it to the required namespace, resource, and verb
- D. Use one shared administrator identity

Q418**ADVANCED**

From a security perspective, a user needs temporary elevated access. What is the strongest response?

- A. Keep broad permissions and review logs later
- B. Retain stale access indefinitely
- C. Use one shared administrator identity
- D. use time-bound elevation with approval and audit

Q419**ADVANCED**

From a security perspective, permissions accumulate after role changes. What is the strongest response?

- A. perform periodic access reviews and remove stale grants
- B. Use one shared administrator identity
- C. Keep broad permissions and review logs later
- D. Retain stale access indefinitely

Q420**EXPERT**

From a security perspective, one shared administrator account is used by the team. What is the strongest response?

- A. Retain stale access indefinitely
- B. Keep broad permissions and review logs later
- C. Use one shared administrator identity
- D. use individual identities and accountable roles

Q421

INTERMEDIATE

From a security perspective, database passwords are stored in a Git repository. What is the strongest response?

- A. remove them, rotate them, and use a secret manager
- B. Base64-encode secrets in source control
- C. Pass secrets on command lines
- D. Reuse one credential across all environments

Q422

ADVANCED

From a security perspective, applications receive secrets as command-line arguments. What is the strongest response?

- A. Pass secrets on command lines
- B. Reuse one credential across all environments
- C. Base64-encode secrets in source control
- D. prefer protected files, descriptors, or injected runtime mechanisms

Q423

ADVANCED

From a security perspective, a secret manager rotates credentials. What is the strongest response?

- A. Pass secrets on command lines
- B. Base64-encode secrets in source control
- C. Reuse one credential across all environments
- D. applications need a safe reload or renewal mechanism

Q424

EXPERT

From a security perspective, one secret is reused across environments. What is the strongest response?

- A. Base64-encode secrets in source control
- B. Pass secrets on command lines
- C. separate credentials to reduce blast radius
- D. Reuse one credential across all environments

Q425

INTERMEDIATE

From a security perspective, an image scanner finds a critical package not used at runtime. What is the strongest response?

- A. A clean vulnerability scan guarantees security
- B. Keep compilers in production images for troubleshooting
- C. validate exploitability but still rebuild or remove unnecessary packages according to policy
- D. Old running images inherit base patches

Q426

ADVANCED

From a security perspective, the base image is months old. What is the strongest response?

- A. Old running images inherit base patches
- B. Keep compilers in production images for troubleshooting
- C. A clean vulnerability scan guarantees security
- D. rebuild regularly against patched base images

Q427

ADVANCED

From a security perspective, a scanner reports no findings. What is the strongest response?

- A. Keep compilers in production images for troubleshooting
- B. the image is not guaranteed secure
- C. A clean vulnerability scan guarantees security
- D. Old running images inherit base patches

Q428

EXPERT

From a security perspective, a production image contains compilers and shells unnecessarily. What is the strongest response?

- A. Keep compilers in production images for troubleshooting
- B. use a minimal runtime image
- C. Old running images inherit base patches
- D. A clean vulnerability scan guarantees security

Q429

INTERMEDIATE

From a security perspective, an artifact signature is valid. What is the strongest response?

- A. Fetch unpinned dependencies
- B. it proves possession of the signing key, not that the artifact is safe
- C. Trust provenance from any worker
- D. A valid signature proves the artifact has no vulnerabilities

Q430

ADVANCED

From a security perspective, a build runs on an untrusted shared worker. What is the strongest response?

- A. A valid signature proves the artifact has no vulnerabilities
- B. Trust provenance from any worker
- C. its provenance should not be trusted for sensitive releases
- D. Fetch unpinned dependencies

Q431

ADVANCED

From a security perspective, dependencies are fetched without hashes. What is the strongest response?

- A. pin versions and verify integrity
- B. Fetch unpinned dependencies
- C. Trust provenance from any worker
- D. A valid signature proves the artifact has no vulnerabilities

Q432

EXPERT

From a security perspective, a package registry account is compromised. What is the strongest response?

- A. revoke credentials, block affected artifacts, and verify downstream exposure
- B. Trust provenance from any worker
- C. A valid signature proves the artifact has no vulnerabilities
- D. Fetch unpinned dependencies

Q433

INTERMEDIATE

From a security perspective, a Kubernetes Role grants get secrets. What is the strongest response?

- A. RoleBinding permissions automatically apply cluster-wide
- B. holders can read secret values in that namespace
- C. Grant cluster-admin for convenience
- D. Secret read access is low risk

Q434

ADVANCED

From a security perspective, a pod can create other pods with arbitrary service accounts. What is the strongest response?

- A. RoleBinding permissions automatically apply cluster-wide
- B. it may escalate privileges
- C. Secret read access is low risk
- D. Grant cluster-admin for convenience

Q435

ADVANCED

From a security perspective, a ClusterRoleBinding is used for a namespaced need. What is the strongest response?

- A. Secret read access is low risk
- B. replace it with a RoleBinding where possible
- C. RoleBinding permissions automatically apply cluster-wide
- D. Grant cluster-admin for convenience

Q436

EXPERT

From a security perspective, admission policy restricts privileged pods. What is the strongest response?

- A. it provides preventive enforcement before creation
- B. Secret read access is low risk
- C. RoleBinding permissions automatically apply cluster-wide
- D. Grant cluster-admin for convenience

Q437

INTERMEDIATE

From a security perspective, SSH password login is exposed to the internet. What is the strongest response?

- A. Accept changed host keys without verification
- B. Share one private key among administrators
- C. Enable direct root password login
- D. use key-based access, MFA or bastion controls, rate limiting, and restricted source networks

Q438

ADVANCED

From a security perspective, a private SSH key is copied among administrators. What is the strongest response?

- A. Accept changed host keys without verification
- B. Enable direct root password login
- C. Share one private key among administrators
- D. issue individual keys and revoke the shared key

Q439

ADVANCED

From a security perspective, a host key changes unexpectedly. What is the strongest response?

- A. Enable direct root password login
- B. Share one private key among administrators
- C. verify through a trusted channel before accepting it
- D. Accept changed host keys without verification

Q440

EXPERT

From a security perspective, root login is enabled directly. What is the strongest response?

- A. use named accounts with sudo and auditability
- B. Share one private key among administrators
- C. Enable direct root password login
- D. Accept changed host keys without verification

Q441

INTERMEDIATE

From a security perspective, a privileged action uses only a password. What is the strongest response?

- A. add phishing-resistant MFA where feasible
- B. Call workload certificates a human second factor
- C. Store recovery codes beside passwords
- D. MFA makes stolen sessions harmless

Q442

ADVANCED

From a security perspective, an attacker steals a session cookie after MFA. What is the strongest response?

- A. MFA does not protect an already hijacked session
- B. Store recovery codes beside passwords
- C. MFA makes stolen sessions harmless
- D. Call workload certificates a human second factor

Q443

ADVANCED

From a security perspective, recovery codes are stored beside the password. What is the strongest response?

- A. separate and protect recovery mechanisms
- B. Call workload certificates a human second factor
- C. Store recovery codes beside passwords
- D. MFA makes stolen sessions harmless

Q444

EXPERT

From a security perspective, service-to-service authentication is called MFA. What is the strongest response?

- A. MFA makes stolen sessions harmless
- B. this is incorrect; workload identity uses different mechanisms
- C. Store recovery codes beside passwords
- D. Call workload certificates a human second factor

Q445

INTERMEDIATE

From a security perspective, a server fetches a URL supplied by the user. What is the strongest response?

- A. Trust every redirect destination
- B. Validate only the original URL text
- C. Allow access to metadata services
- D. validate destinations and block access to internal metadata and private ranges

Q446

ADVANCED

From a security perspective, URL validation checks only the original hostname. What is the strongest response?

- A. Allow access to metadata services
- B. Validate only the original URL text
- C. Trust every redirect destination
- D. DNS rebinding or redirects can bypass it

Q447

ADVANCED

From a security perspective, cloud metadata is reachable without protection. What is the strongest response?

- A. Allow access to metadata services
- B. Validate only the original URL text
- C. use metadata service hardening and network controls
- D. Trust every redirect destination

Q448

EXPERT

From a security perspective, an allowlist includes arbitrary subdomains. What is the strongest response?

- A. Validate only the original URL text
- B. Trust every redirect destination
- C. Allow access to metadata services
- D. consider takeover and DNS control risks

Q449

INTERMEDIATE

From a security perspective, SQL is built by concatenating user input. What is the strongest response?

- A. Give the application database-owner privileges
- B. use parameterized queries
- C. Escape SQL manually in every code path
- D. Rely only on a WAF

Q450

ADVANCED

From a security perspective, input is escaped manually. What is the strongest response?

- A. prefer database-driver parameters
- B. Give the application database-owner privileges
- C. Rely only on a WAF
- D. Escape SQL manually in every code path

Q451

ADVANCED

From a security perspective, the database account can drop every schema. What is the strongest response?

- A. restrict permissions needed by the application
- B. Escape SQL manually in every code path
- C. Give the application database-owner privileges
- D. Rely only on a WAF

Q452

EXPERT

From a security perspective, a WAF blocks common payloads. What is the strongest response?

- A. Rely only on a WAF
- B. Escape SQL manually in every code path
- C. the application still needs safe query construction
- D. Give the application database-owner privileges

Q453

INTERMEDIATE

From a security perspective, a container runs with --privileged. What is the strongest response?

- A. Disable seccomp
- B. Run privileged instead of adding one capability
- C. Keep a writable root filesystem when unnecessary
- D. it receives extensive host capabilities and device access

Q454

ADVANCED

From a security perspective, an app needs no Linux capabilities. What is the strongest response?

- A. Disable seccomp
- B. Run privileged instead of adding one capability
- C. drop all and add back only required ones
- D. Keep a writable root filesystem when unnecessary

Q455

ADVANCED

From a security perspective, the root filesystem can be read-only. What is the strongest response?

- A. Disable seccomp
- B. Run privileged instead of adding one capability
- C. enable it and provide writable mounts only where needed
- D. Keep a writable root filesystem when unnecessary

Q456

EXPERT

From a security perspective, seccomp is disabled. What is the strongest response?

- A. Disable seccomp
- B. Keep a writable root filesystem when unnecessary
- C. the process can invoke a broader syscall set
- D. Run privileged instead of adding one capability

Q457

INTERMEDIATE

From a security perspective, a database subnet accepts traffic from the entire internet. What is the strongest response?

- A. Trust all east-west traffic
- B. Expose database subnets publicly
- C. Keep development and production on one unrestricted network
- D. restrict it to required application sources

Q458

ADVANCED

From a security perspective, development and production share flat network access. What is the strongest response?

- A. Expose database subnets publicly
- B. Keep development and production on one unrestricted network
- C. separate environments and control interconnections
- D. Trust all east-west traffic

Q459

ADVANCED

From a security perspective, east-west traffic is implicitly trusted. What is the strongest response?

- A. Keep development and production on one unrestricted network
- B. apply service identity and network policy based on risk
- C. Expose database subnets publicly
- D. Trust all east-west traffic

Q460

EXPERT

From a security perspective, a firewall rule permits any port temporarily. What is the strongest response?

- A. Expose database subnets publicly
- B. time-limit, document, monitor, and remove it promptly
- C. Keep development and production on one unrestricted network
- D. Trust all east-west traffic

Q461

INTERMEDIATE

From a security perspective, a critical internet-facing vulnerability has active exploitation. What is the strongest response?

- A. Patch only assets already known informally
- B. Delay an actively exploited patch until the next annual cycle
- C. prioritize emergency mitigation or patching
- D. Assume package installation updates running processes

Q462

ADVANCED

From a security perspective, a patch requires downtime. What is the strongest response?

- A. Delay an actively exploited patch until the next annual cycle
- B. Patch only assets already known informally
- C. use redundancy, rolling maintenance, or a controlled window
- D. Assume package installation updates running processes

Q463

ADVANCED

From a security perspective, a package is patched but the service was not restarted. What is the strongest response?

- A. Delay an actively exploited patch until the next annual cycle
- B. Assume package installation updates running processes
- C. Patch only assets already known informally
- D. the old vulnerable code may still be loaded

Q464

EXPERT

From a security perspective, inventory is incomplete. What is the strongest response?

- A. Assume package installation updates running processes
- B. Delay an actively exploited patch until the next annual cycle
- C. patch management cannot reliably cover unknown assets
- D. Patch only assets already known informally

Q465

INTERMEDIATE

From a security perspective, a production credential is suspected stolen. What is the strongest response?

- A. Wait for full root cause before revoking a stolen credential
- B. revoke or rotate it immediately and investigate use
- C. End the incident process immediately after service restoration
- D. Destroy all evidence while isolating a host

Q466**ADVANCED**

From a security perspective, malware is active on one host. What is the strongest response?

- A. Wait for full root cause before revoking a stolen credential
- B. isolate the host while preserving evidence
- C. Destroy all evidence while isolating a host
- D. End the incident process immediately after service restoration

Q467**ADVANCED**

From a security perspective, an incident channel includes secrets copied from logs. What is the strongest response?

- A. remove exposure and use secure evidence handling
- B. End the incident process immediately after service restoration
- C. Wait for full root cause before revoking a stolen credential
- D. Destroy all evidence while isolating a host

Q468**EXPERT**

From a security perspective, service restoration is complete. What is the strongest response?

- A. continue eradication, recovery validation, and lessons learned
- B. Destroy all evidence while isolating a host
- C. End the incident process immediately after service restoration
- D. Wait for full root cause before revoking a stolen credential

Architecture and Incident Response

ARCHITECTURE AND INCIDENT RESPONSE

Questions 469-520 | 52 questions

Architecture and Incident Response Questions

Q469**INTERMEDIATE**

For system architecture or incident response, a producer spikes faster than a consumer can process. What is the best decision?

- A. buffer work in a durable queue and apply backpressure
- B. Retry producers indefinitely without backpressure
- C. Assume a queue solves permanent under-capacity
- D. Process messages with non-idempotent side effects

Q470**ADVANCED**

For system architecture or incident response, the queue grows continuously. What is the best decision?

- A. Assume a queue solves permanent under-capacity
- B. Process messages with non-idempotent side effects
- C. consumer capacity or processing health is insufficient
- D. Retry producers indefinitely without backpressure

Q471**ADVANCED**

For system architecture or incident response, a message can be processed twice. What is the best decision?

- A. make side effects idempotent
- B. Retry producers indefinitely without backpressure
- C. Process messages with non-idempotent side effects
- D. Assume a queue solves permanent under-capacity

Q472**EXPERT**

For system architecture or incident response, a queue is the only copy of critical data. What is the best decision?

- A. Assume a queue solves permanent under-capacity
- B. Process messages with non-idempotent side effects
- C. Retry producers indefinitely without backpressure
- D. configure durability, replication, and recovery according to the RPO

Q473

INTERMEDIATE

For system architecture or incident response, a popular cache key expires and thousands of requests hit the database. What is the best decision?

- A. Treat cache data as automatically durable
- B. use request coalescing, jittered TTLs, or stale serving
- C. Ignore stale-data semantics after writes
- D. Expire every hot key at the same time

Q474

ADVANCED

For system architecture or incident response, a cache stores negative lookup results briefly. What is the best decision?

- A. Treat cache data as automatically durable
- B. this can reduce repeated misses
- C. Expire every hot key at the same time
- D. Ignore stale-data semantics after writes

Q475

ADVANCED

For system architecture or incident response, cache invalidation is delayed after writes. What is the best decision?

- A. Expire every hot key at the same time
- B. Treat cache data as automatically durable
- C. users may observe stale data
- D. Ignore stale-data semantics after writes

Q476

EXPERT

For system architecture or incident response, the cache is treated as the system of record. What is the best decision?

- A. Expire every hot key at the same time
- B. Ignore stale-data semantics after writes
- C. data loss can occur unless it is designed as durable storage
- D. Treat cache data as automatically durable

Q477

INTERMEDIATE

For system architecture or incident response, a downstream dependency is failing and callers retry continuously. What is the best decision?

- A. open a circuit breaker and fail fast
- B. Let every layer create an unbounded fallback chain
- C. Retry a failed dependency continuously
- D. Keep the breaker open forever with no probes

Q478

ADVANCED

For system architecture or incident response, the breaker is half-open. What is the best decision?

- A. allow limited probes to test recovery
- B. Keep the breaker open forever with no probes
- C. Retry a failed dependency continuously
- D. Let every layer create an unbounded fallback chain

Q479

ADVANCED

For system architecture or incident response, every instance opens its breaker independently. What is the best decision?

- A. Keep the breaker open forever with no probes
- B. Let every layer create an unbounded fallback chain
- C. Retry a failed dependency continuously
- D. recovery traffic may still surge

Q480

EXPERT

For system architecture or incident response, fallback data is stale but acceptable. What is the best decision?

- A. Keep the breaker open forever with no probes
- B. Retry a failed dependency continuously
- C. Let every layer create an unbounded fallback chain
- D. graceful fallback can preserve partial functionality

Q481

INTERMEDIATE

For system architecture or incident response, a payment request is retried after a timeout. What is the best decision?

- A. use an idempotency key
- B. Retry non-idempotent operations blindly
- C. Delete deduplication records immediately
- D. Acknowledge work before durable side effects

Q482

ADVANCED

For system architecture or incident response, a create operation is naturally non-idempotent. What is the best decision?

- A. Acknowledge work before durable side effects
- B. Retry non-idempotent operations blindly
- C. design a stable client token or deduplication record
- D. Delete deduplication records immediately

Q483

ADVANCED

For system architecture or incident response, a consumer marks work complete before committing the result. What is the best decision?

- A. Delete deduplication records immediately
- B. Retry non-idempotent operations blindly
- C. Acknowledge work before durable side effects
- D. a crash can lose the side effect

Q484

EXPERT

For system architecture or incident response, deduplication records expire too soon. What is the best decision?

- A. Retry non-idempotent operations blindly
- B. Delete deduplication records immediately
- C. late retries can repeat effects
- D. Acknowledge work before durable side effects

Q485

INTERMEDIATE

For system architecture or incident response, a read immediately after a write must return the new value. What is the best decision?

- A. Use one data model for every consistency need
- B. Require the strongest consistency without accepting latency tradeoffs
- C. Assume replicas are always current
- D. provide read-your-writes consistency

Q486

ADVANCED

For system architecture or incident response, replicas lag behind the primary. What is the best decision?

- A. eventual reads may return stale data
- B. Use one data model for every consistency need
- C. Assume replicas are always current
- D. Require the strongest consistency without accepting latency tradeoffs

Q487

ADVANCED

For system architecture or incident response, all operations require linearizability. What is the best decision?

- A. Use one data model for every consistency need
- B. Assume replicas are always current
- C. latency and availability tradeoffs increase
- D. Require the strongest consistency without accepting latency tradeoffs

Q488

EXPERT

For system architecture or incident response, a counter is updated concurrently in several regions. What is the best decision?

- A. Assume replicas are always current
- B. use a design suited to conflict resolution or centralized coordination
- C. Use one data model for every consistency need
- D. Require the strongest consistency without accepting latency tradeoffs

Q489

INTERMEDIATE

For system architecture or incident response, a network partition occurs in a distributed system. What is the best decision?

- A. Ignore partitions in distributed design
- B. Apply CAP directly to a single local process
- C. CAP means a system can never be both consistent and available
- D. the system must trade some availability against strong consistency

Q490

ADVANCED

For system architecture or incident response, a design claims consistency, availability, and partition tolerance with no tradeoff. What is the best decision?

- A. Ignore partitions in distributed design
- B. the claim is incomplete for partition scenarios
- C. CAP means a system can never be both consistent and available
- D. Apply CAP directly to a single local process

Q491**ADVANCED**

For system architecture or incident response, a single-node database is discussed under CAP. What is the best decision?

- A. Apply CAP directly to a single local process
- B. Ignore partitions in distributed design
- C. CAP means a system can never be both consistent and available
- D. the larger practical issue may be availability rather than distributed partition tradeoffs

Q492**EXPERT**

For system architecture or incident response, different operations choose different consistency levels. What is the best decision?

- A. Ignore partitions in distributed design
- B. Apply CAP directly to a single local process
- C. the system can make per-operation tradeoffs
- D. CAP means a system can never be both consistent and available

Q493**INTERMEDIATE**

For system architecture or incident response, a user-facing outage affects all customers. What is the best decision?

- A. Stop stakeholder updates until resolution
- B. Allow all responders to make uncoordinated changes
- C. Debate complete root cause before mitigation
- D. declare a high-severity incident and establish command roles

Q494**ADVANCED**

For system architecture or incident response, engineers debate root cause while impact continues. What is the best decision?

- A. Stop stakeholder updates until resolution
- B. prioritize mitigation and containment first
- C. Allow all responders to make uncoordinated changes
- D. Debate complete root cause before mitigation

Q495

ADVANCED

For system architecture or incident response, multiple responders make conflicting changes. What is the best decision?

- A. assign an incident commander and change coordination
- B. Debate complete root cause before mitigation
- C. Allow all responders to make uncoordinated changes
- D. Stop stakeholder updates until resolution

Q496

EXPERT

For system architecture or incident response, status updates stop during a long incident. What is the best decision?

- A. Stop stakeholder updates until resolution
- B. Allow all responders to make uncoordinated changes
- C. maintain a communication cadence with known facts and uncertainty
- D. Debate complete root cause before mitigation

Q497

INTERMEDIATE

For system architecture or incident response, a postmortem names one engineer as the cause. What is the best decision?

- A. Mix assumptions into the factual timeline
- B. Assign blame to the last operator
- C. shift toward systemic contributing factors and safeguards
- D. Create actions without owners

Q498

ADVANCED

For system architecture or incident response, the document lists actions without owners or dates. What is the best decision?

- A. Mix assumptions into the factual timeline
- B. Create actions without owners
- C. assign accountable owners and due dates
- D. Assign blame to the last operator

Q499**ADVANCED**

For system architecture or incident response, the same incident repeats. What is the best decision?

- A. Assign blame to the last operator
- B. verify whether prior actions were completed and effective
- C. Mix assumptions into the factual timeline
- D. Create actions without owners

Q500**EXPERT**

For system architecture or incident response, the timeline uses guesses as facts. What is the best decision?

- A. Mix assumptions into the factual timeline
- B. Assign blame to the last operator
- C. separate evidence, assumptions, and unresolved questions
- D. Create actions without owners

Q501**INTERMEDIATE**

For system architecture or incident response, average CPU is 30% but peak traffic saturates every evening. What is the best decision?

- A. Plan only from annual average utilization
- B. Assume autoscaling removes all capacity planning
- C. plan for peaks, headroom, and failure scenarios
- D. Ignore N-1 failure capacity

Q502**ADVANCED**

For system architecture or incident response, one AZ fails and remaining AZs cannot carry traffic. What is the best decision?

- A. Ignore N-1 failure capacity
- B. Assume autoscaling removes all capacity planning
- C. Plan only from annual average utilization
- D. the service lacks failure-capacity headroom

Q503**ADVANCED**

For system architecture or incident response, load tests use unrealistic requests. What is the best decision?

- A. their capacity conclusions are unreliable
- B. Plan only from annual average utilization
- C. Assume autoscaling removes all capacity planning
- D. Ignore N-1 failure capacity

Q504**EXPERT**

For system architecture or incident response, growth is 10% monthly. What is the best decision?

- A. Ignore N-1 failure capacity
- B. Assume autoscaling removes all capacity planning
- C. Plan only from annual average utilization
- D. forecast compounding demand and lead times

Q505**INTERMEDIATE**

For system architecture or incident response, one client can consume all API capacity. What is the best decision?

- A. Multiply a per-instance limit and call it global
- B. Allow one client to consume all capacity
- C. Retry 429 responses immediately
- D. apply fair rate limits and quotas

Q506**ADVANCED**

For system architecture or incident response, a client receives 429. What is the best decision?

- A. Retry 429 responses immediately
- B. Allow one client to consume all capacity
- C. Multiply a per-instance limit and call it global
- D. it should honor Retry-After and back off

Q507**ADVANCED**

For system architecture or incident response, limits are enforced independently on every replica. What is the best decision?

- A. Retry 429 responses immediately
- B. Multiply a per-instance limit and call it global
- C. Allow one client to consume all capacity
- D. the effective global limit may multiply with scale

Q508**EXPERT**

For system architecture or incident response, critical internal traffic competes with bulk jobs. What is the best decision?

- A. Allow one client to consume all capacity
- B. use priority classes or separate capacity pools
- C. Retry 429 responses immediately
- D. Multiply a per-instance limit and call it global

Q509

INTERMEDIATE

For system architecture or incident response, retries occur immediately and synchronously across thousands of clients. What is the best decision?

- A. Use immediate synchronized retries
- B. use exponential backoff with jitter
- C. Retry non-idempotent work without a key
- D. Let every layer retry independently

Q510

ADVANCED

For system architecture or incident response, an operation is non-idempotent. What is the best decision?

- A. do not retry blindly
- B. Use immediate synchronized retries
- C. Retry non-idempotent work without a key
- D. Let every layer retry independently

Q511

ADVANCED

For system architecture or incident response, every layer retries three times. What is the best decision?

- A. Use immediate synchronized retries
- B. total attempts can multiply dramatically
- C. Let every layer retry independently
- D. Retry non-idempotent work without a key

Q512

EXPERT

For system architecture or incident response, the timeout is longer than the caller's deadline. What is the best decision?

- A. the downstream work may continue after the caller has abandoned it
- B. Let every layer retry independently
- C. Use immediate synchronized retries
- D. Retry non-idempotent work without a key

Q513

INTERMEDIATE

For system architecture or incident response, the recommendation engine fails but checkout can operate. What is the best decision?

- A. Hide stale fallback data when it affects decisions
- B. Fail checkout when recommendations fail
- C. Use unbounded dependency timeouts
- D. disable recommendations and preserve checkout

Q514

ADVANCED

For system architecture or incident response, a dependency times out slowly. What is the best decision?

- A. Use unbounded dependency timeouts
- B. use bounded timeouts and fallback behavior
- C. Fail checkout when recommendations fail
- D. Hide stale fallback data when it affects decisions

Q515

ADVANCED

For system architecture or incident response, fallback responses look identical to fresh data. What is the best decision?

- A. Fail checkout when recommendations fail
- B. indicate staleness when it matters to user decisions
- C. Use unbounded dependency timeouts
- D. Hide stale fallback data when it affects decisions

Q516

EXPERT

For system architecture or incident response, every feature depends synchronously on one personalization service. What is the best decision?

- A. Use unbounded dependency timeouts
- B. Fail checkout when recommendations fail
- C. Hide stale fallback data when it affects decisions
- D. the architecture has a broad failure coupling

Q517

INTERMEDIATE

For system architecture or incident response, backups exist but restores have never been tested. What is the best decision?

- A. Depend on one expert for disaster failover
- B. Keep obsolete runbooks
- C. the recovery capability is unproven
- D. Treat successful backup jobs as proven recovery

Q518**ADVANCED**

For system architecture or incident response, a regional failover runbook references obsolete resources. What is the best decision?

- A. update and exercise it
- B. Keep obsolete runbooks
- C. Depend on one expert for disaster failover
- D. Treat successful backup jobs as proven recovery

Q519**ADVANCED**

For system architecture or incident response, DNS failover TTL is one hour but RTO is ten minutes. What is the best decision?

- A. the design may not meet the RTO
- B. Depend on one expert for disaster failover
- C. Keep obsolete runbooks
- D. Treat successful backup jobs as proven recovery

Q520**EXPERT**

For system architecture or incident response, a disaster exercise succeeds only with one expert present. What is the best decision?

- A. Depend on one expert for disaster failover
- B. Treat successful backup jobs as proven recovery
- C. reduce key-person dependency through automation and training
- D. Keep obsolete runbooks

Answer Key and Explanations

The answer key begins here. Each explanation identifies the principle being tested; review missed questions by domain rather than memorizing letters.

Answer Key - Linux

Question	Answer	Explanation
Q001	C	High I/O wait and D-state tasks show processes blocked on I/O.
Q002	D	A high blocked-task count and heavy block input indicate storage waits, not CPU saturation.
Q003	C	Load average includes tasks waiting in uninterruptible sleep, including NFS waits.
Q004	D	High load with low CPU after a SAN issue strongly indicates I/O-bound blocked tasks.
Q005	C	MemAvailable estimates memory usable without heavy swapping; page cache is reclaimable.
Q006	D	Used memory includes cache that can be reclaimed when applications need RAM.
Q007	C	Low MemAvailable plus major faults is evidence of pressure, unlike low MemFree alone.
Q008	D	Sustained si/so activity means pages are moving between RAM and swap.
Q009	B	A zombie has exited but remains in the process table until its parent calls wait().
Q010	A	The parent is failing to reap children; killing individual zombies does not solve the cause.
Q011	B	A zombie has no running code to signal; the parent must reap it.
Q012	D	The init process becomes the parent of orphans and is responsible for reaping them.
Q013	C	A filesystem can have free bytes but no free inodes.
Q014	C	Each file consumes an inode; freeing inodes requires deleting files or recreating the filesystem with more inodes.
Q015	C	Small files can consume all inodes while leaving significant byte capacity.
Q016	D	The immediate fix is cleanup; the durable fix is lifecycle management.
Q017	D	A regular script needs an execute bit when invoked directly.
Q018	B	Directory execute permission controls traversal and access to entries.
Q019	D	setgid on a directory causes new entries to inherit the directory group.
Q020	D	A targeted capability is safer than full root privilege.
Q021	D	The unit journal normally contains the actionable startup error.
Q022	B	systemd does not inherit an interactive shell environment.
Q023	A	network.target does not guarantee usable connectivity.
Q024	B	systemd must track the real main process correctly.
Q025	C	SIGTERM can be handled so the application can clean up.
Q026	A	SIGKILL cannot be caught, but it prevents cleanup.
Q027	D	Many daemons use SIGHUP to reload configuration.
Q028	C	exec replaces the shell, allowing signals to reach the application directly.
Q029	A	The error usually means RLIMIT_NOFILE or the system-wide file table is exhausted.
Q030	B	Interactive shell limits may not affect systemd services.
Q031	C	CLOSE_WAIT persists until the local application closes the socket.
Q032	C	Raising limits only delays a true descriptor leak.
Q033	D	Deleted files continue consuming space until the owning process closes them.
Q034	C	Unlinking removes the directory entry, not the open file handle.
Q035	D	copytruncate has a race between copying and truncation.
Q036	C	Reopening closes the old descriptor and releases disk space safely.
Q037	B	Cron runs with a minimal environment.
Q038	D	Cron's current directory is not guaranteed to match the script directory.
Q039	D	Non-interactive cron jobs do not reliably source user profiles.
Q040	A	The five cron fields are minute, hour, day of month, month, day of week.
Q041	D	ss can show listening TCP sockets and owning processes.
Q042	A	Loopback bindings are reachable only from the local host.

Question	Answer	Explanation
Q043	B	The failure is on the destination host before a TCP session is established.
Q044	B	Local success only proves the process is running, not that the path is open.
Q045	A	Both the block device and filesystem must be expanded.
Q046	A	The LV can grow without the ext4 filesystem automatically growing.
Q047	D	XFS grows online using xfs_growfs and does not shrink.
Q048	A	Shrinking the LV first can destroy data; XFS cannot be shrunk in place.
Q049	B	Steal time is CPU time the guest wanted but the hypervisor assigned elsewhere.
Q050	C	High steal points outside the guest to oversubscribed physical CPU.
Q051	B	The guest has runnable work but is waiting for physical CPU.
Q052	C	Guest tuning cannot fully solve hypervisor-level contention.

Answer Key - Git

Question	Answer	Explanation
Q053	D	Reflog records recent reference movements and usually allows recovery before expiration.
Q054	C	Reflog is the fastest way to find the previous branch tip.
Q055	B	Unreachable objects survive until pruning and can be rescued by a new branch or tag.
Q056	D	Deleting a branch removes the name, not immediately the commit objects.
Q057	D	A rebase rewrites history; force-with-lease protects against overwriting unseen remote work.
Q058	B	The lease verifies the remote still matches your expected value.
Q059	A	It adds a safety check against unexpected remote updates.
Q060	D	Rewriting public history forces all collaborators to reconcile divergent commits.
Q061	A	Rebase replays feature commits on the new base.
Q062	B	Merge preserves existing commit identities and branch history.
Q063	A	The merge commit preserves the integration event.
Q064	D	Interactive rebase is appropriate for cleaning unpublished history.
Q065	C	New commits are not attached to a movable branch name unless you create one.
Q066	D	A branch preserves the detached commits with a durable reference.
Q067	B	Tags identify commits but do not move like branches.
Q068	A	A branch provides a safe place for new commits.
Q069	B	Bisect performs a binary search through commit history.
Q070	D	Automation can narrow the faulty commit efficiently.
Q071	D	Binary search takes about $\log_2(N)$ steps.
Q072	C	Reset returns the working tree to the original branch and state.
Q073	B	Revert adds a new commit that inverses the old change without rewriting history.
Q074	A	Git needs to know which parent represents the line to keep.
Q075	A	Reset would rewrite branch history and disrupt collaborators.
Q076	C	Reverting the revert reapplies the original logical change with a new commit.
Q077	C	Stash saves tracked modifications and optionally untracked files.
Q078	C	The -u option includes untracked files.
Q079	D	Apply leaves the stash entry in the list; pop removes it after applying.
Q080	C	Stash application can conflict just like a merge.
Q081	D	Cherry-pick applies the selected commit's change onto the current branch.
Q082	B	The sequencer waits for resolved files before continuing.
Q083	C	Abort returns to the pre-cherry-pick state.
Q084	B	Ranges can apply multiple selected commits without merging the whole branch.
Q085	B	Staging marks the conflict as resolved.
Q086	D	Git could not automatically merge the content.
Q087	C	The labels ours/theirs depend on the operation and should not replace review.
Q088	C	Git cannot text-merge arbitrary binary content.
Q089	B	Annotated signed tags contain metadata and a cryptographic signature.
Q090	B	Lightweight tags lack annotation metadata and a tag object.
Q091	D	Normal branch pushes do not necessarily push tags.
Q092	C	Retagging breaks reproducibility for consumers who already fetched the old tag.
Q093	D	A submodule entry points to a commit, not automatically the latest branch tip.
Q094	C	Submodule content must be initialized and checked out separately.

Question	Answer	Explanation
Q095	A	The parent must record the new referenced commit.
Q096	D	Submodules add operational complexity and are not always the best dependency mechanism.
Q097	C	An upstream enables concise pull, push, and status comparisons.
Q098	A	Pull combines fetch with an integration strategy that should be explicit.
Q099	B	Fetch downloads refs and objects but does not merge them.
Q100	A	Pruning removes stale remote-tracking references.
Q101	A	Git's object model is inefficient for frequently changing large binaries.
Q102	A	The secret remains in prior commits and should be treated as compromised.
Q103	B	Garbage collection optimizes storage and may eventually remove unreachable objects.
Q104	B	Generated artifacts should usually not be versioned when reproducible.

Answer Key - CI/CD

Question	Answer	Explanation
Q105	C	Immutable identifiers make each deployment reproducible.
Q106	B	Artifact promotion prevents environment-specific rebuild drift.
Q107	C	A digest identifies the exact image content.
Q108	B	Build-once promotion reduces supply-chain and reproducibility risk.
Q109	A	Short-lived scoped credentials reduce exposure and rotation burden.
Q110	C	Masking cannot protect a value deliberately echoed into logs.
Q111	C	Least privilege includes limiting where and when a secret is available.
Q112	A	Isolation reduces the chance of untrusted jobs accessing privileged credentials.
Q113	D	Parallelism reduces cycle time without weakening gates.
Q114	C	A deployment should consume only validated artifacts.
Q115	C	A precise key improves speed without reusing incompatible dependencies.
Q116	D	The DAG should block consumers when required producers fail.
Q117	B	Artifacts transfer immutable outputs between stages.
Q118	D	Caches are performance optimizations; artifacts are authoritative outputs.
Q119	B	Untrusted jobs must not populate caches consumed by privileged jobs.
Q120	C	Operational recovery depends on retaining deployable versions.
Q121	C	Canaries limit initial blast radius and provide real traffic validation.
Q122	D	Traffic can move between full old and new stacks quickly.
Q123	A	Compatibility is required while old and new versions overlap.
Q124	B	Canary analysis should include latency, saturation, and business signals.
Q125	A	Fast rollback reduces impact when validation is objective.
Q126	A	Application rollback cannot restore destroyed data automatically.
Q127	D	Untested recovery procedures are unreliable during incidents.
Q128	D	Progressive delivery should contain blast radius by stage.
Q129	C	Ignoring flaky tests erodes trust in the pipeline.
Q130	A	Retries can gather evidence but also hide defects.
Q131	A	Shared state creates ordering and concurrency failures.
Q132	D	Optimization should preserve coverage while reducing feedback time.
Q133	C	Independent work can reduce wall-clock time through concurrency.
Q134	C	Parallelism must respect downstream capacity and state isolation.
Q135	B	Balanced work minimizes the critical path.
Q136	D	Serialization is required for shared mutable deployment targets.
Q137	C	A disclosed credential should be treated as compromised.
Q138	D	Secret masking is not a substitute for disciplined logging.
Q139	A	Remote unverified execution is a supply-chain risk.
Q140	C	Untrusted code could exfiltrate them.
Q141	C	A gate turns policy into an enforceable release condition.
Q142	B	Risk-based severity and exploitability should drive gating.
Q143	C	Quality tooling must remain actionable.
Q144	D	Automation provides consistent enforcement and auditability.
Q145	B	The gate records and restricts who can authorize production.
Q146	A	Exceptions should be controlled, visible, and reviewed.

Question	Answer	Explanation
Q147	B	Independent approval reduces fraud and error risk.
Q148	D	Approvers should review the exact artifact that will deploy.
Q149	B	An SBOM inventories packages and versions for risk analysis.
Q150	C	Signatures provide provenance when keys and policy are managed correctly.
Q151	C	Existing images do not change when a base tag is updated.
Q152	D	Attestations link source, builder, process, and output.
Q153	D	A successful command does not prove the service works for users.
Q154	C	Shallow health endpoints can miss functional failures.
Q155	D	Release automation must account for telemetry latency.
Q156	B	Controllers roll out when desired state changes.

Answer Key - Docker

Question	Answer	Explanation
Q157	A	The host-side bind address limits where the published port listens.
Q158	A	A 0.0.0.0 or specific external bind can accept remote traffic subject to firewall policy.
Q159	C	The process must listen on the container interface, typically 0.0.0.0.
Q160	A	Runtime publishing is controlled by -p or orchestration.
Q161	C	The writable layer follows the container lifecycle.
Q162	D	Persistent data should live outside the ephemeral writable layer.
Q163	C	Bind mounts expose a specific host path directly.
Q164	A	The -v flag explicitly deletes associated volumes.
Q165	C	A builder stage can compile while the final stage contains only runtime files.
Q166	A	Smaller images reduce attack surface and transfer time.
Q167	C	ARG values can leak through image history and layers.
Q168	A	Layer ordering improves cache reuse.
Q169	B	The build cache is ordered by layer dependency.
Q170	C	Combining avoids stale package indexes and unnecessary layer data.
Q171	D	Compiled outputs and package caches may be platform-specific.
Q172	A	Excluding irrelevant files reduces context transfer and accidental inclusion.
Q173	C	Process existence alone does not prove application health.
Q174	C	Startup tolerance prevents premature failure classification.
Q175	B	Health checks should not create their own overload or dependency cascades.
Q176	B	Replacement behavior depends on an orchestrator or external policy.
Q177	B	The application then receives signals directly as PID 1.
Q178	D	Containers still need child-process reaping.
Q179	B	The application should handle the configured stop signal promptly.
Q180	D	Orderly termination reduces dropped requests and corruption.
Q181	D	Least privilege reduces impact if the process is compromised.
Q182	B	Full root is unnecessary for one narrow privilege.
Q183	B	The Docker API can start privileged containers and mount host files.
Q184	C	Rootless mode reduces daemon-level privilege but has feature tradeoffs.
Q185	C	Tags are mutable references; digests are content-addressed.
Q186	B	A digest validates content identity.
Q187	D	A digest prevents silent tag movement.
Q188	B	The top-level digest can select different child manifests by platform.
Q189	A	Layers are additive; deletion creates a whiteout rather than rewriting old layers.
Q190	C	Cleanup must occur in the same layer that creates the data.
Q191	D	Persistent volumes avoid some writable-layer overhead and lifecycle coupling.
Q192	A	Prune should be used carefully on hosts that rely on stopped containers or unused images.
Q193	A	CPU limits generally constrain scheduling rather than kill the process.
Q194	A	Memory cannot be throttled in the same way as CPU.
Q195	D	Default isolation does not guarantee a safe upper bound.
Q196	C	Runtime awareness is required to size the heap correctly.
Q197	C	User-defined networks support name-based discovery.
Q198	A	Each container normally has its own network namespace.

Question	Answer	Explanation
Q199	B	Multiple interfaces can bridge selected communication paths.
Q200	B	User-defined networks provide better DNS and isolation.
Q201	C	A started container may not yet be ready to serve.
Q202	A	Distributed startup requires tolerance for delayed dependencies.
Q203	A	Startup ordering does not solve runtime failures.
Q204	B	Runtime secret injection avoids image-layer exposure.
Q205	C	Unbounded local logs can fill the host filesystem.
Q206	A	Container platforms can capture standard streams consistently.
Q207	D	Log transport security cannot undo sensitive content already emitted.
Q208	D	Blocking logging can stall applications; dropping logs reduces observability.

Answer Key - Kubernetes

Question	Answer	Explanation
Q209	A	A selector mismatch leaves the Service with no backend endpoints.
Q210	B	Only selected, eligible Pods become service endpoints.
Q211	D	The Service maps its port to the target port.
Q212	B	Headless Services support direct endpoint discovery.
Q213	A	Startup probes suppress liveness and readiness until startup succeeds.
Q214	B	Readiness controls endpoint eligibility, not process restart.
Q215	C	Liveness should usually test whether the local process can recover by restart.
Q216	A	Probe settings must reflect realistic healthy behavior.
Q217	C	No old Pod may be removed first and no new Pod can schedule.
Q218	B	The strategy permits temporary unavailability of one replica.
Q219	B	Pause supports batching changes before rollout.
Q220	C	The controller identifies a stalled rollout but does not automatically roll back.
Q221	A	CPU quota limits execution time.
Q222	B	Memory limit violations can terminate processes.
Q223	C	Requests drive scheduling and QoS.
Q224	B	Guaranteed requires CPU and memory requests equal limits for every container.
Q225	C	Priority does not override hard affinity or taints.
Q226	A	NoSchedule blocks new Pods without matching toleration.
Q227	B	Hard affinity is a scheduling requirement.
Q228	D	Preferred rules influence scoring but are not mandatory.
Q229	D	Binding depends on compatible storage and provisioning.
Q230	C	Access modes describe node-level attachment semantics.
Q231	B	StatefulSet identity includes per-replica storage.
Q232	C	Delayed binding accounts for topology.
Q233	A	CrashLoopBackOff is a restart backoff state, not the root cause.
Q234	D	A controller restarts the successfully exited container.
Q235	B	The kernel terminated the process due to memory pressure or limit violation.
Q236	D	Backoff reduces rapid restart churn.
Q237	C	The event stream usually identifies auth, DNS, or not-found errors.
Q238	B	Secrets are namespace-scoped.
Q239	B	Immutable tags or digests avoid cache ambiguity.
Q240	C	The image manifest must support the node architecture.
Q241	C	The network path to the service works, isolating the problem to DNS.
Q242	B	service resolves first within the caller's namespace.
Q243	D	Default changes normal Kubernetes service-name resolution behavior.
Q244	D	Resolver behavior can generate multiple queries before an absolute lookup.
Q245	B	PDBs protect availability during voluntary disruptions.
Q246	B	PDBs guide eviction, not hardware failure.
Q247	D	There is no spare replica to preserve availability.
Q248	C	PDBs govern eviction, not workload desired replica count.
Q249	D	Utilization is measured relative to requested CPU.
Q250	B	One clear owner should control the replica target.

Question	Answer	Explanation
Q251	B	Autoscaling has detection and startup delay.
Q252	A	Heap retention can make memory a poor demand signal.
Q253	C	Authentication succeeded but authorization denied the verb/resource.
Q254	C	RoleBinding scope remains namespaced.
Q255	A	Least privilege avoids broad cluster permissions.
Q256	B	This helps validate RBAC decisions.
Q257	A	A default 404 usually means no matching rule reached the backend.
Q258	C	The controller matched the route but could not proxy successfully.
Q259	B	A gateway timeout indicates the request did not complete in time.
Q260	D	IngressClass should make controller ownership explicit.

Answer Key - Cloud

Question	Answer	Explanation
Q261	D	The multi-AZ application tier still depends on one failure domain.
Q262	D	Restore time remains downtime and recent changes may be lost.
Q263	D	A standby can assume service without a full restore.
Q264	B	Multi-AZ and multi-region address different failure scopes.
Q265	D	True resilience requires all critical dependencies to avoid the failed zone.
Q266	D	Nominal multi-AZ configuration is insufficient without balanced healthy capacity.
Q267	A	Compute movement alone does not move zonal state.
Q268	B	Resilience decisions include operational tradeoffs.
Q269	A	Reactive CPU signals can lag fast traffic changes.
Q270	C	Autoscaling cannot help immediately if provisioning is slow.
Q271	A	Scaling one tier can overload a constrained dependency.
Q272	D	Graceful removal prevents dropped work.
Q273	D	Durability and availability are distinct service characteristics.
Q274	B	Previous object versions remain available subject to lifecycle policy.
Q275	C	Guardrails reduce exposure from individual misconfiguration.
Q276	D	Cloud cost depends on operations as well as capacity.
Q277	D	Edge caching serves content closer to users.
Q278	D	The key must vary on all relevant request attributes.
Q279	B	These techniques reduce cache stampedes.
Q280	C	Signing helps authorize private content delivery.
Q281	B	Least privilege reduces impact of compromised credentials.
Q282	A	The trust policy is as important as the permission policy.
Q283	A	Access analysis supports privilege reduction.
Q284	B	Long-lived human keys are difficult to govern safely.
Q285	A	RPO is the acceptable data-loss window.
Q286	D	RTO is the recovery-time objective.
Q287	B	Protection frequency is too low for the required data-loss window.
Q288	A	Objectives must hold under realistic incident conditions.
Q289	A	Predictable usage can justify reserved capacity.
Q290	B	Interruptible pricing fits restartable workloads.
Q291	C	The workload is neither redundant nor interruption-tolerant.
Q292	C	Optimization should precede long-term commitment.
Q293	A	Cloud billing often charges for inter-region transfer.
Q294	C	Locality can reduce repeated cross-region traffic.
Q295	C	Managed NAT often charges per processed byte.
Q296	B	Telemetry volume can create material transfer expense.
Q297	B	Queues decouple processing rates and absorb bursts.
Q298	D	Many managed queues provide at-least-once delivery.
Q299	A	Global order is expensive and often unnecessary.
Q300	D	Isolation enables inspection without blocking the whole stream.
Q301	B	NAT enables egress while instances remain privately addressed.
Q302	C	Private endpoints keep traffic on provider networks and bypass NAT processing.

Question	Answer	Explanation
Q303	C	Address overlap requires redesign, translation, or an intermediary architecture.
Q304	B	The route alone does not make private addresses internet-routable.
Q305	C	Platform execution limits shape workload design.
Q306	B	Initialization cost varies by runtime and configuration.
Q307	B	Burst concurrency can exhaust database connection limits.
Q308	B	At-least-once invocation can repeat side effects.
Q309	C	KMS authorization is part of the data access path.
Q310	A	Disabling a key can make protected data temporarily unavailable.
Q311	B	Rotation should preserve access to existing ciphertext.
Q312	A	Key boundaries support isolation and independent revocation.

Answer Key - Monitoring and Observability

Question	Answer	Explanation
Q313	C	An absolute rate ignores service traffic volume.
Q314	A	20 divided by 10,000 equals 0.002.
Q315	C	Ratio alerts can be noisy at tiny denominators.
Q316	A	Global averages can mask localized failures.
Q317	B	rate handles counter resets and normalizes by time.
Q318	C	increase estimates total counter growth in the window.
Q319	A	irate uses only the last two samples and is noisier.
Q320	C	Raw subtraction would appear negative.
Q321	A	Unbounded identifiers create one time series per value.
Q322	C	Templated routes avoid unbounded series growth.
Q323	A	Series count is a major resource driver.
Q324	B	Metrics should use bounded dimensions.
Q325	D	Server-side buckets allow aggregation across instances.
Q326	A	Precomputed quantiles do not combine mathematically.
Q327	D	Quantile accuracy depends on bucket placement.
Q328	B	Metrics intentionally summarize many observations.
Q329	D	Non-actionable alerts train responders to ignore notifications.
Q330	D	Paging should be reserved for time-sensitive action.
Q331	C	Noise obscures the primary failure.
Q332	B	Actionability requires clear responsibility and diagnosis steps.
Q333	A	An SLI is a quantitative measure of service behavior.
Q334	C	An SLO sets the desired level for an SLI.
Q335	A	An SLA is a formal agreement with consequences.
Q336	C	Good SLIs reflect outcomes that matter to users.
Q337	C	Rapid budget consumption threatens the SLO immediately.
Q338	A	Multi-window burn rates detect both fast and slow problems.
Q339	D	Error budgets balance reliability and change velocity.
Q340	C	The budget provides a decision mechanism, not just a report.
Q341	A	Structured events improve search, parsing, and correlation.
Q342	B	A shared correlation identifier links events.
Q343	A	Sensitive data should never be emitted.
Q344	D	Volume and privacy costs can outweigh diagnostic value.
Q345	C	Context links producer and consumer spans.
Q346	D	Sampling policy affects diagnostic coverage.
Q347	D	Critical path spans explain end-to-end duration.
Q348	A	Cross-signal correlation speeds diagnosis.
Q349	D	A scrape failure can occur anywhere along the collection path.
Q350	D	The up metric reflects scrape success, not application availability.
Q351	B	Expensive exporters can destabilize monitoring.
Q352	D	Prometheus cannot scrape a target it no longer discovers.
Q353	C	Counters start again from zero when the process restarts.
Q354	B	Rates are additive across independent counters.

Question	Answer	Explanation
Q355	C	Raw totals embed process lifetime and resets.
Q356	C	Counters represent cumulative non-decreasing events.
Q357	B	Precomputed series reduce repeated query cost.
Q358	A	Centralized calculations reduce copy-paste drift.
Q359	A	Evaluation cadence should match the signal.
Q360	C	Precomputation does not remove label-design responsibility.
Q361	B	It observes the service from a user's external perspective.
Q362	A	Internal telemetry explains component state.
Q363	D	External dependencies may fail outside the application.
Q364	D	Black-box signals detect symptoms but may not explain cause.

Answer Key - Networking

Question	Answer	Explanation
Q365	D	TCP connection establishment uses SYN, SYN-ACK, ACK.
Q366	C	No SYN-ACK means connection establishment never completed.
Q367	B	A reset is different from a silent timeout.
Q368	D	The network path and listener worked at least through establishment.
Q369	C	NXDOMAIN is a negative name response.
Q370	D	DNSSEC, upstream, timeout, or authoritative problems can cause SERVFAIL.
Q371	B	Resolvers can legally retain the old value until TTL expiration.
Q372	A	Different views can intentionally serve different records.
Q373	D	Bad Gateway generally means the proxy could not use the upstream response.
Q374	B	Gateway Timeout indicates an upstream latency or reachability delay.
Q375	A	503 can signal temporary capacity or maintenance conditions.
Q376	C	499 is an NGINX-specific client-closed code.
Q377	B	The requested host must match a SAN on the certificate.
Q378	C	Servers should present required intermediate certificates.
Q379	D	Modern servers may disable obsolete algorithms.
Q380	C	SNI selects the virtual host during the TLS handshake.
Q381	C	A /24 leaves eight host bits.
Q382	B	The /25 lies inside the /24.
Q383	D	Network and broadcast consume two of four addresses in conventional subnets.
Q384	B	All 32 bits are fixed.
Q385	B	Encapsulation can make large packets exceed path capacity.
Q386	C	Endpoints may keep sending packets too large for the path.
Q387	C	MSS adjustment accounts for encapsulation overhead.
Q388	D	End-to-end support is required.
Q389	B	Port translation distinguishes simultaneous sessions.
Q390	B	Outbound source NAT alone does not publish the service.
Q391	D	Stateful translation must retain the flow mapping.
Q392	A	Some protocols require helpers or application awareness.
Q393	C	Application-aware routing requires HTTP visibility.
Q394	B	L4 balances transport connections without parsing HTTP.
Q395	B	Termination changes the encryption boundary.
Q396	C	NAT-based balancing can replace the apparent client address.
Q397	C	TIME_WAIT preserves sequence-space safety.
Q398	B	The application must close its side.
Q399	A	This can result from load, packet loss, or SYN flooding.
Q400	A	Connection state alone does not guarantee progress.
Q401	B	Routing chooses the most specific matching prefix.
Q402	B	Administrative policy can select among equal-length routes.
Q403	B	A host needs a matching route for outbound forwarding.
Q404	C	Stateful devices expect both directions of a flow.
Q405	B	Many firewall chains process rules in order and stop at the first terminal match.
Q406	D	State tracking allows reply traffic efficiently.

Question	Answer	Explanation
Q407	B	Firewall changes need safe deployment controls.
Q408	C	Counters help validate actual rule paths.
Q409	B	Mismatched idle timeouts cause unexpected disconnects.
Q410	D	Reuse reduces handshake overhead.
Q411	C	Aggressive probes create traffic and false positives.
Q412	B	Timeout budgets must reflect expected processing.
Q413	C	Packet capture shows whether traffic reaches and leaves the host.
Q414	B	The inbound path works to the interface; response generation is local.
Q415	B	Capture point matters when interpreting duplicates.
Q416	A	Encryption protects application content.

Answer Key - Security

Question	Answer	Explanation
Q417	C	Least privilege limits damage from misuse or compromise.
Q418	D	Standing privilege increases exposure.
Q419	A	Privilege creep is a common long-term risk.
Q420	D	Shared credentials destroy attribution and complicate revocation.
Q421	A	Deleting the latest file does not remove historical exposure.
Q422	D	Command lines may be visible in process listings and logs.
Q423	D	Rotation is useful only if consumers can adopt new values.
Q424	C	Environment isolation supports independent revocation and auditing.
Q425	C	Context matters, but reducing unused components lowers risk.
Q426	D	Running images do not inherit later base-image fixes.
Q427	B	Scanners have database, coverage, configuration, and zero-day limits.
Q428	B	Smaller attack surface reduces available tools and vulnerable packages.
Q429	B	Trust also depends on key control, provenance, and policy.
Q430	C	The builder is part of the supply chain.
Q431	A	Unpinned remote content can change unexpectedly.
Q432	A	Supply-chain incidents require containment across consumers.
Q433	B	Secret access is highly sensitive even without write permission.
Q434	B	Workload creation can be an authorization escalation path.
Q435	B	Cluster-wide binding broadens effective scope.
Q436	A	Admission controls complement RBAC by validating object properties.
Q437	D	Layered controls reduce brute-force and credential risk.
Q438	D	Shared private keys cannot provide accountability.
Q439	C	An unexpected change can indicate rebuild or man-in-the-middle attack.
Q440	A	Direct root access reduces attribution and control.
Q441	A	A second independent factor reduces credential-only compromise.
Q442	A	Session security and reauthentication remain necessary.
Q443	A	Co-located factors can fail together.
Q444	B	MFA is designed around human authentication factors.
Q445	D	SSRF can turn the server into a network proxy.
Q446	D	Validation must cover resolved addresses and redirect chains.
Q447	C	Metadata credentials are a frequent SSRF target.
Q448	D	Trusting a namespace requires confidence in every allowed host.
Q449	B	Parameters separate data from executable SQL syntax.
Q450	A	Manual escaping is error-prone and context-dependent.
Q451	A	Least privilege limits injection impact.
Q452	C	A WAF is a compensating control, not a complete fix.
Q453	D	Privileged containers greatly weaken isolation.
Q454	C	Capability minimization reduces kernel attack surface.
Q455	C	Immutability limits persistence and tampering.
Q456	C	Seccomp filters reduce exposed kernel interfaces.
Q457	D	Segmentation limits reachable attack paths.
Q458	C	Isolation reduces accidental and malicious lateral movement.

Question	Answer	Explanation
Q459	B	Internal location alone is not strong authentication.
Q460	B	Temporary broad access often becomes permanent risk.
Q461	C	Exploitability and exposure drive urgency.
Q462	C	Operational planning should not justify indefinite vulnerability.
Q463	D	Many updates require process restart or reboot.
Q464	C	Asset visibility is foundational to vulnerability management.
Q465	B	Containment should stop continued access before full root-cause analysis completes.
Q466	B	Containment reduces spread without unnecessarily destroying forensic data.
Q467	A	Incident urgency does not remove data-protection requirements.
Q468	A	Recovery is one phase of the incident lifecycle.

Answer Key - Architecture and Incident Response

Question	Answer	Explanation
Q469	A	Queues decouple rates and protect downstream services.
Q470	C	A queue absorbs bursts but cannot solve permanent under-capacity.
Q471	A	At-least-once systems require duplicate-safe consumers.
Q472	D	The broker itself is part of the data-protection design.
Q473	B	These patterns prevent a cache stampede.
Q474	B	Negative caching protects backends from nonexistent-key storms.
Q475	C	The consistency model must be explicit.
Q476	C	Most caches prioritize speed over durability.
Q477	A	Circuit breaking prevents repeated load on an unhealthy dependency.
Q478	A	Half-open state evaluates whether normal traffic can resume.
Q479	D	Distributed breakers need jitter, limits, and coordinated capacity awareness.
Q480	D	Fallback quality must be explicit to users and operators.
Q481	A	The server can recognize and return the prior result instead of charging twice.
Q482	C	Retries are unavoidable in distributed systems.
Q483	D	Ordering and transactional boundaries determine delivery guarantees.
Q484	C	Retention should cover the maximum retry window.
Q485	D	The same user should observe its completed update.
Q486	A	Replication delay affects consistency.
Q487	C	Strong consistency has coordination cost.
Q488	B	The required semantics determine the data model.
Q489	D	CAP concerns behavior during partitions.
Q490	B	Under partition, every request cannot always have both strong consistency and availability.
Q491	D	CAP is often misapplied outside distributed replication.
Q492	C	Real systems often combine models rather than choose one globally.
Q493	D	Broad critical impact requires coordinated response.
Q494	B	Restoring service is usually more urgent than complete diagnosis.
Q495	A	Clear authority reduces accidental escalation.
Q496	C	Stakeholders need reliable operational visibility.
Q497	C	Blameless analysis improves learning and reporting.
Q498	C	Unowned lessons rarely become improvements.
Q499	B	Action closure should include evidence of risk reduction.
Q500	C	Accurate timelines support credible learning.
Q501	C	Averages hide burst and redundancy requirements.
Q502	D	Resilience requires N-1 capacity or rapid scaling.
Q503	A	Workload mix and dependency behavior must resemble production.
Q504	D	Capacity changes may require procurement or quota approval.
Q505	D	Rate limiting protects shared resources.
Q506	D	Immediate retries can worsen overload.
Q507	D	Distributed enforcement needs shared state or intentional per-node semantics.
Q508	B	Not all traffic has equal business importance.
Q509	B	Jitter prevents synchronized retry storms.
Q510	A	Retries can duplicate side effects.

Question	Answer	Explanation
Q511	B	Retry ownership and budgets should be coordinated.
Q512	A	Propagate deadlines and cancellation where possible.
Q513	D	Graceful degradation protects core business functions.
Q514	B	Slow failure can exhaust threads and connection pools.
Q515	B	Degradation should not mislead users.
Q516	D	Optional features should not block the critical path.
Q517	C	Backup success does not guarantee restorable data or acceptable timing.
Q518	A	Documentation decays as systems change.
Q519	A	Client caching can delay traffic movement.
Q520	C	Recovery must be repeatable by the on-call organization.

End of Question Bank

Revisit weak domains, then retake a mixed set under time pressure. Aim to explain why each distractor is wrong, not only why the correct option is right.